

Adhesion to Enamel and Dentin

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Adhesion

The process of *adhesion* may be defined simply as the joining together of two independent surfaces or materials for which contact is maintained without the aid of external forces. This definition does not say anything about the type, relative strength, or long-term stability of the adhesive joint but only declares that it exists. In the strictest sense, *adhesion* is a measure of the force of attraction between two different materials and is differentiated from *cohesion* in that the latter relates to the forces of attraction within a single material, ie, the forces holding it together. Adhesion can result from the formation of primary chemical bonds, such as covalent, ionic, or metallic, and strong, durable joints can be produced from each of these types of bonds. Adhesion also can result from strong secondary forces, such as hydrogen bonds and Van der Waals forces, which typically are of lower overall

durability than primary chemical bonds because of their lower bonding energies. Whether based on primary or secondary bonds, adhesion depends on strong molecular interactions between two surfaces in intimate contact, and both types of interactions are evident in dental adhesive materials. It is also common to "lock" one surface or material into another through a purely mechanical interaction, mediated by frictional forces between surfaces in close contact. This mechanical bonding does not truly represent an adhesive joint, as it depends on undercuts and obstructions to motion for its stability.

The extent to which one surface or material interacts with another depends on many factors, the most important relating to energy.[1] In nature, there is a desire for all objects to seek a reduced energy state, simply because this is the most stable condition. Surfaces, in general, are of higher energy than the internal aspects of an object because molecules present at the surface have unsatisfied bonds. In other words, these molecules would prefer to be "covered" by other molecules to satisfy their bond complexes and reduce their overall energy state. This covering can occur by oxygen, water, or other molecules. The higher the energy of the surface, the more receptive it is to being bonded to by another material, such as an adhesive. Thus, practical approaches to raising the energy of a surface are typically sought to enhance adhesion. In dentistry, such approaches include surface cleaning by pumice or prophylactic pastes, etching with mild or strong acids, or cleaning with solvents to remove contaminants. When an adhesive material, typically in a liquid or paste form to facilitate application, is applied to a surface for bonding, the adhesive will to some extent spread out on the surface. This relative spreading is referred to as *wetting*, and it is very important to be able to enhance this physical effect in order to ensure good bonding. The reason is fairly obvious if one considers that bond strength is affected to a great extent by how much of the two independent materials are brought into intimate contact with one another. Thus, a liquid that beads up on a surface, like water on a freshly waxed automobile, does not spread and provides a minimum of surfaces in contact for bonding (Fig 9-1).

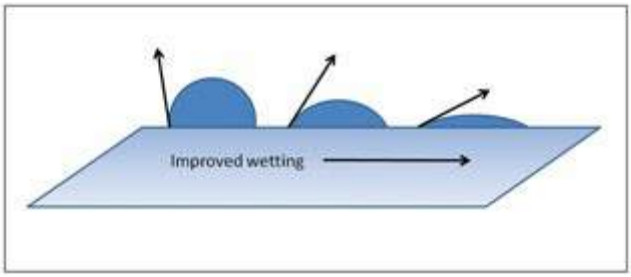


Fig 9-1 The *contact angle*, the angle between the surface and a tangent line drawn to the edge of the adhesive droplet through the droplet itself, depicts how well the adhesive wets the surface. The lower the contact angle, the better an adhesive will wet the surface and enhance adhesion.

On the other hand, solder flowing out onto a metal surface to create a union between two clean metal parts provides an example of good wetting. One critical property of the adhesive that affects the amount of spreading is the surface tension of the adhesive itself. The molecules of the adhesive have an affinity for one another that causes them to stay together rather than interact with the surface they contact. If this surface energy of the adhesive, or *surface tension*, is very high, the adhesive tends to bead up and not spread. Thus, an ideal condition for bonding is to have a substrate with high surface energy (ie, one that is clean) and an adhesive with low surface tension (ie, one that does not bead up).

The chemistry of the adhesive determines how well it will wet the substrate and ultimately how strongly it will attach. Like molecules attract like molecules, and therefore two materials with similar surface chemistry are more likely to interact and adhere. These interactions may come in the form of chemical interactions, such as primary chemical bonds, or via polar or secondary chemical interactions, such as hydrogen bonds. As noted previously, the latter are typically of lower bond energy and considered less stable. However, there are numerous examples of strong and durable bonds formed primarily from these secondary forces, such as wood glues and bonding to acid-etched enamel. In both cases, strong bonds are formed by an adhesive with relatively low surface tension flowing into all of the microscopic irregularities of a surface and spreading across it, providing intimate contact and the possibility for molecular interactions. These same requirements are present for establishing primary chemical bonds as well. Therefore, it is easy to understand why anything that inhibits the close contact between an adhesive and a surface, measured on the order of atomic sizes, will inhibit wetting and the molecular interactions required for adhesion.

Critical variables influencing adhesion are the surface roughness of the substrate, its cleanliness (ie, its lack of contamination), the viscosity of the adhesive, the dimensional change occurring in the adhesive during setting, and the durability of the adhesive and the new interface.[2] A rough surface, especially one with microscopic roughness, may produce a very high surface area and enhanced potential for surface interactions, as long as the adhesive can spread onto the surface and flow into the irregularities. This is the case with acid etching of enamel for bonding dental resin materials. This is also why the viscosity of the adhesive is so important; if the adhesive is too viscous, it will not flow adequately and cannot displace the air and/or molecules already present on the surface in order to create the required intimate contact. If the adhesive is too fluid, it will not stay in place on the substrate. The need for low viscosity to enhance flow explains why most adhesives or primers for

creating bonding to dentin have solvents to reduce their viscosity. However, if the surface has already been contaminated by molecules that have an equal or greater affinity for the surface, they will not be displaced by the adhesive, and the intimate contact will be hindered and the bonding reduced. While it is possible that the adhesive has a high affinity for interacting with the contaminant surface, an unstable situation may be created as the adhesive bonds to the contaminant because the stability of the overall joint then depends on how strongly the contaminant is attached to the substrate. This explains the need for cleaning the surface to produce a known, clean substrate for bonding. Frequent contaminants of tooth surfaces, such as water, saliva, and blood, must be routinely managed by the practitioner, typically with good isolation techniques such as rubber dam, cotton rolls, or other devices.

Once the adhesive is applied to the surface, it must transform from a liquid to a solid to effect bonding. Most dental adhesives, based on polymeric materials, undergo significant shrinkage during this hardening process. This shrinkage may have a deleterious effect on the bond by creating forces that physically pull the adhesive away from the substrate. This is especially a concern when the adhesive material is simultaneously being bonded to two opposing and rigid substrates, like the opposite walls of a cavity preparation. One can imagine this "tug of war" as the adhesion is being created between the two sides at nearly the same time that the material is pulling away by virtue of setting shrinkage. This issue is addressed to a great extent in later chapters (see chapter 11). Finally, the interface that forms between the substrate and the adhesive is expected to be durable and to resist external physical forces as well as chemical or biologic degradation, or else the joint fails. This complex interplay as it relates to bonding to dentin is addressed later in this chapter.

Substrates

The composition of the enamel and dentin substrates has been thoroughly reviewed in chapter 1. In terms of their influence on achieving adhesion, the important aspects of these substrates include their abundance of mineral as a potential site for chemical interactions in both and how the adhesion is affected by the high concentration of organic material in dentin, specifically collagen, as well as the significant water content of the latter. In addition, both substrates are highly *hydrophilic*, ie, water loving. It may seem that developing an adhesive for bonding to both of these substrates at the same time would be impossible, but the development of adhesive primers with enhanced hydrophilicity as agents for preparing the surfaces for bonding to the more *hydrophobic* (ie, water hating or resin loving) resin adhesive

has largely solved this concern, albeit after many years of study and development.

In the oral cavity, certain conditions may affect the normality of the enamel and dentin substrates and potentially influence bonding. Enamel and dentin may become hypomineralized, either through caries attack or as a direct result of a developmental anomaly, such as amelogenesis imperfecta, or through erosion, such as may occur with a patient suffering from bulimia or gastroesophageal reflux disease (GERD). These surfaces may be more difficult to bond to because of the weakened nature of the substrate.[3–6] In contrast, exposure to mineralizing solutions, such as fluoride, via intraoral application or via the water supply in certain regions, may produce hypermineralized enamel. This latter material may be more resistant to demineralization and potentially requires a longer etching time to produce an adequate surface for bonding. Teeth that have been highly stained from systemic absorption of chemicals in drug therapies, such as tetracycline, also may provide a modified enamel surface for bonding.

Over time, dentin may become more *sclerotic* (ie, calcified), thus affecting how well it may be demineralized during normal adhesive placement. Laboratory studies have shown that bonding to sclerotic dentin may be reduced as compared with normal dentin[7] but that extending etching time may enhance the bond for certain adhesives using strong acids.[8] The bond to sclerotic dentin is also lower than that for normal dentin when using adhesives containing milder organic acids (such as those used in the self-etching adhesives described later in this chapter),[9,10] but additional tooth preparation with burs or air abrasion may be beneficial for enhancing the bond.[11] Despite this reduced bond to sclerotic dentin, at least one clinical study showed equivalent results at 8 years for Class 5 restorations bonded to sclerotic and nonsclerotic dentin.[12] Others have reported clinical outcomes that show enhanced performance of the adhesives containing stronger acids when dealing with sclerotic dentin.[13]

Materials

Dental adhesives consist of three main components: *(1)* etchant, *(2)* primer, and *(3)* bonding resin. The latter is often referred to as the *adhesive resin*, but the entire system also is typically called an *adhesive*. The etchant is composed of acidic molecules that alter or remove the smear layer (defined in the Adhesive System Classification section) and demineralize the enamel and dentin and prepare it for bonding. The primer serves as a type of molecule that helps make the dentin surface, which is very hydrophilic, become more hydrophobic in order to accept the very

hydrophobic bonding resin. Thus, the primer typically contains a molecule or molecules with both hydrophilic and hydrophobic characteristics, ie, *amphiphilic*. It functions by penetrating the demineralized dentin and preparing it for the bonding resin. The bonding resin then becomes incorporated into the primed dentin and, once cured, forms the structural support of the bonded interface between the tooth and the subsequently placed restorative material. The entire process forms a zone of material that consists of components of both the dentin and the resin and is called the *hybrid layer* or *interdiffusion zone*.

All dental adhesive systems are based on polymeric materials. These materials have a variety of chemical characteristics, ranging from very hydrophilic (glass ionomer) to very hydrophobic (the adhesive resin component of many adhesive systems). Considering the hydrophilic nature of the tooth, especially dentin, it is logical that an adhesive also should have a hydrophilic nature when placed in order to wet the surface and penetrate the available microstructure. Attempts to bond unfilled monomers, the same as those used in dental composites and sealants, to an etched dentin surface is unsuccessful in large part because of the lack of hydrophilicity of the materials.[14] While a detriment during placement, this hydrophobic nature is considered to be beneficial once the material has established a bond with the tooth substrate. A hydrophobic polymeric layer is more insoluble and resistant to erosion and degradation by acids and other components of oral fluids than a more hydrophilic one.[15,16] Thus, the ideal adhesive would have a hydrophilic nature during placement and would become much more hydrophobic after curing. Such materials are not readily available to dentistry—instead a combination of hydrophilic and hydrophobic molecules are used in modern adhesives to effect durable bonding, with varied results explained in greater detail later in this chapter. Because enamel contains little water, hydrophobic monomers are capable of wetting and penetrating its etched structure with ease, as long as it has been dried adequately.

The major difference between hydrophilic and hydrophobic adhesives is the chemistry of their monomers and solvents. Monomers with alcohol, acid, hydroxyl, and amino groups are more capable of enhancing chemical interactions with the collagen and hydroxyapatite of tooth structure than are molecules consisting predominantly of hydrocarbons. As hydroxyl, carboxylic, phosphate, and amide functional groups are substituted for groups containing hydrogen, methyl, ethyl, and so forth, on the polymer chain, and as ether and ester linkages are incorporated, the polymer becomes slightly more hydrophilic (Table 9-1). However, *hydrophilicity* is a relative term. A monomer like hydroxylethyl methacrylate (HEMA) is totally

miscible in water and serves as an excellent polymerizable wetting agent for dental adhesives (see Table 9-1). Bisphenol glycidyl methacrylate (bis-GMA), the main monomer used in most dental composites and many adhesives (the final component of the bonding system application process), is much more hydrophobic and will only absorb about 3% water by weight into its structure when polymerized[17] (see Table 9-1). A mixture of the two has intermediate characteristics and serves as a useful adhesive for the tooth. To enhance the wetting, spreading, and penetration of the polymerizable monomers into the dentin substrate, solvents are always added to the mixture as "thinning" agents. These solvents are typically water, ethyl alcohol, butyl alcohol, or acetone. The first three are very hydrophilic and thus enhance the interaction of the monomers with surface water, while acetone is good at displacing ("drying") water from within the dentin. However, any solvent not displaced during the placement procedure, such as by drying appropriately,[18] will be incorporated into the bonding layer and may serve as a weakening contaminant. Because it is not a polymerizable molecule and cannot bond with the newly formed polymer, such an agent is called a *plasticizer*, so named because it softens the polymer structure in the bonding layer by expanding the polymer network and reducing the beneficial physical entanglements of individual polymer chains that contribute to its strength.

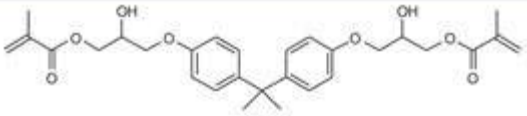
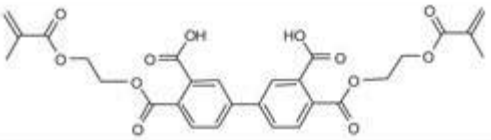
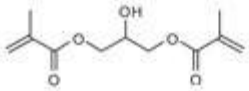
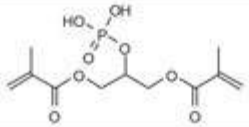
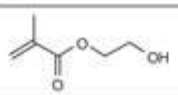
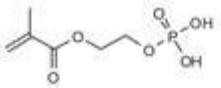
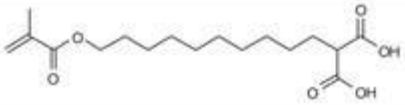
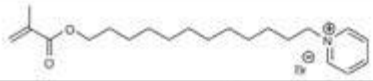
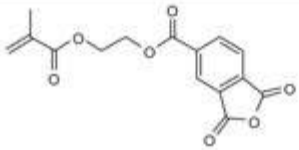
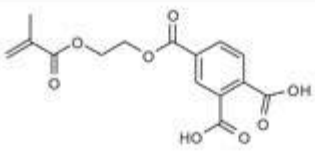
Table 9-1	Abbreviations and molecular formulas of monomers, initiators and inhibitors, and additives and coupling agents used in adhesives
Abbreviation	**Molecular formula**
Monomers	
Bis-GMA bisphenol A glycerolate dimethacrylate *or* 2,2-bis (p-2'-hydroxy-3'-methacryloxypropoxyphenyl)-propane	
BPDM biphenyl dimethacrylate *or* 4,4-dimethacryloyloxyethyloxy-carbonylbiphenyl-3,3-dicarboxylic acid	
GDMA glycerol dimethacrylate *or* glycerol-1,3-dimethacrylate	
GPDM glycerol phosphate dimethacrylate *or* glycerol-2-dihydrogenphosphate-1,3-dimethacrylate	
HDDMA 1,6-hexanediol dimethacrylate	
HEMA 2-hydroxyethyl methacrylate	
HEMA-phosphate 2-hydroxyethyl methacryl dihydrogenphosphate *or* 2-dihydrogenphosphate-1-methacrylate-etane	
MAC-10 11-methacryloyloxy-1,1'-undecanedicarboxylic acid *or* 2-carboxyl-12-methacryl-dodecanoic acid	
10-MDP 10-methacryloyloxydecyl dihydrogenphosphate *or* 10-methacrylate-1-dihydrogenphosphate-decane	
MDPB methacryloyloxydodecylpyridinium bromide *or* 12-methacryl-1-pyridyl-dodecane bromide	
4-META 4-methacryloxyethyl trimellitate anhydride *or* 4-methacryloxy-ethyl trimellitic anhydride	
4-MET 4-methacryloyloxyethyl trimellitic acid	

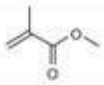
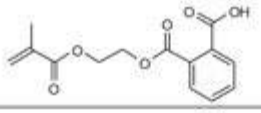
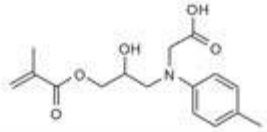
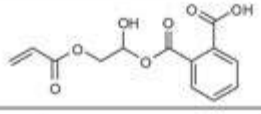
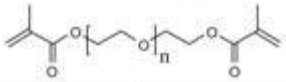
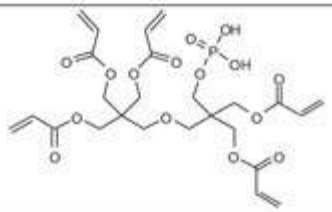
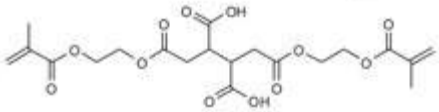
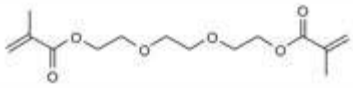
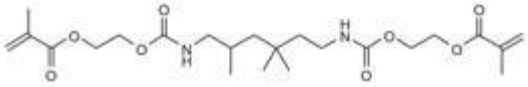
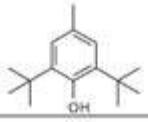
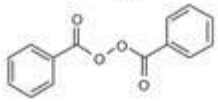
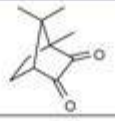
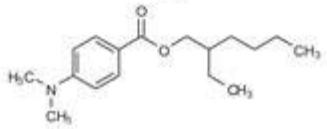
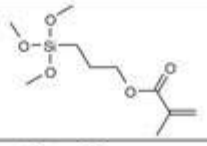
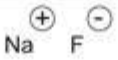
Table 9-1 *(cont)*	Abbreviations and molecular formulas of monomers, initiators and inhibitors, and additives and coupling agents used in adhesives
Abbreviation	**Molecular formula**
Monomers	
MMA methyl methacrylate	
MMEP mono-2-methacryloyloxyethyl phthalate *or* PAMA: phthalic acid monomethacrylate	
NGT-GMA N-tolylglycine glycidyl methacrylate *or* N-(2-hydroxy-3-[(2-methyl-1-oxo-2-propenyl)oxy]propyl)-N-tolyl glycine	
PAMM mono-(2-methacryloxyethyl)-phthalate *or* methacryloxyethyl phthalate *or* phthalic acid monoethyl methacrylate	
PEGDMA polyethylene glycol dimethacrylate	
PENTA dipentaerythritol pentaacrylate monophosphate	
TCB butan-1,2,3,4-tetracarboxylic acid di-2'-hydroxyethyl-1'-methacrylate ester	
TEGDMA triethylenglycol dimethacrylate	
UDMA urethane dimethacrylate *or* 1,6-di(methacryloyloxyethylcarbamoyl)-3,3,5-trimethylhexaan	
Initiators and inhibitors	
BHT butylhydroxytoluene *or* butylated hydroxytoluene *or* ,2,6-di-(tert-butyl)-4-methylphenol	
BPO benzoylperoxide (redox initiator)	

Table 9-1 *(cont)*	Abbreviations and molecular formulas of monomers, initiators and inhibitors, and additives and coupling agents used in adhesives
Abbreviation	**Molecular formula**
Initiators and inhibitors	
CQ camphorquinone *or* 1,7,7-trimethylbicyclo-[2,2,1]-hepta-2,3-dione (photoinitiator)	
EDMAB 2-(ethylhexyl)-4-(dimethylamino)benzoate (co-initiator)	
Additives and silane coupling agents	
Coupling agent A174 methacryloxypropyltrimethoxysilane	
NaF sodium fluoride	

The monomers present in dental adhesives are similar to those used in dental composite restoratives, thus ensuring that there will be strong interaction between the adhesive and the overlying composite (see Table 9-1). But critical to achieving bonding to the tooth is the conversion of the monomers that penetrated the tooth structure into a rigid, strong polymer. This process, called *polymerization*, involves the chemical splitting of carbon-carbon double bonds of the monomers, followed by the formation of carbon-carbon single bonds linking one monomer to another, thus building a long chain of molecules much like adding links to a chain (Fig 9-2).

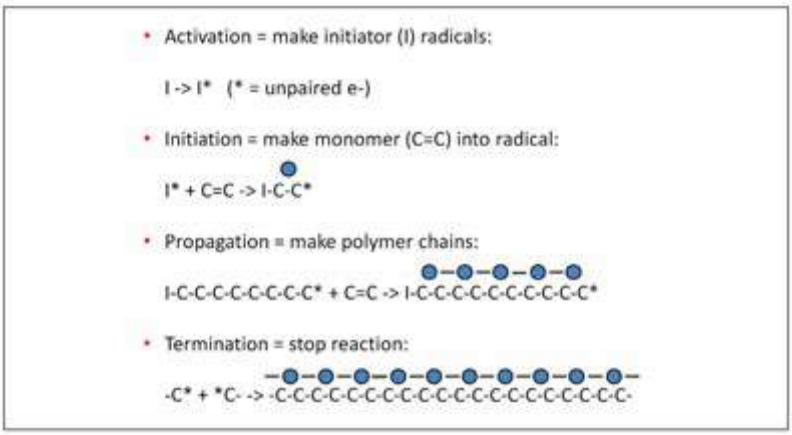


Fig 9-2 The steps in the process of the free radical polymerization reaction of an adhesive, showing the activation step that produces a free radical on the initiator (I); the initiation step where the active initiator (I*) collides with a monomer (C=C) and makes it become active (C*); the propagation step where the monomer containing the active radical (C-C*) collides sequentially with additional monomers, transferring the radical and extending the polymer chain; and the termination step where the radicals are extinguished and the polymer chain is terminated.

In fact, the reaction is termed an *addition reaction* for this reason. The covalent bonds linking molecules together are strong and produce a relatively stiff bonding layer that is capable of resisting many types of mechanical and chemical forces. The polymerization process typically is begun by exposing the adhesive to radiation from a high-intensity light source tuned to the same wavelength range (400 to 500 nm, blue light) that is absorbed by the purposely added photoinitiator.[19] Thus, photons of blue light are directed into the uncured adhesive, and the photoinitiator molecule within (typically camphorquinone) absorbs these photons, raising these molecules to an excited state from which they are capable of interacting chemically with the monomer molecules to split carbon-carbon double bonds and initiate the polymerization reaction. It is also possible to use chemicals in the absence of light energy to produce the reaction in what are called *self-curing materials*, as opposed to the *light-curing materials* just described. The extent to which the monomer molecules react to produce polymer determines the length of the polymer chains, which is directly related to the strength of the final polymer.[20] Thus, exposing the adhesive to adequate light energy and effecting adequate polymerization is critical for forming a strong and durable bonded interface with the tooth.

Accompanying the polymerization reaction is a volumetric change of the adhesive material. This is not surprising, because the molecules that formerly interacted at the level of secondary forces within a liquid state have now become permanently linked by stronger, primary chemical bonds characterized by shorter distances between molecules. The overall volume contraction depends on the size of the molecules reacting and the extent to which they reacted (polymerized) and can be as high as 10% for an unfilled resin. The same effect occurs when the dental composite, composed of much the same chemicals, polymerizes to fill the cavity preparation, forming the restoration. However, as the composite is typically filled heavily with reinforcing glass particles, its shrinkage is much lower because much of its volume is taken up by the nonshrinking filler. This difference in shrinkage between dental composites (particulate-filled polymers) and unfilled resins, as well as the reinforcing effect of the filler particles, explains in part why manufacturers often incorporate fillers in their adhesives. The goal is to decrease polymerization shrinkage and increase strength, while taking care not to add so much filler as to detrimentally increase the viscosity of the mixture, inhibiting its ability to flow and penetrate the surface. The shrinkage of the material during curing produces a dimensional change that causes stresses to build up between the tooth surface and the adhesive and ultimately between the tooth and the composite restorative through the adhesive.[21] These stresses can be very detrimental in that they may lead to the

composite and adhesive being pulled away from the tooth surface to which it is trying to establish a bond (Fig 9-3).

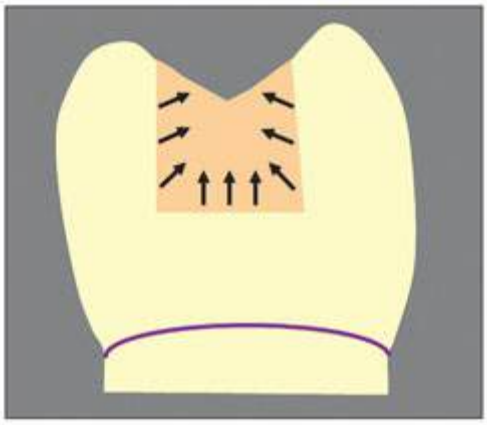


Fig 9-3 Curing of the dental composite causes it to shrink, pulling the material away from the cavity walls of the tooth and toward its center.

As noted earlier, the ideal adhesive would not change dimension during curing to avoid this potential problem. This explains why there is so much development of low-shrinking monomers for use in dental adhesives, composites, and cements.

Once the bond to the tooth is established, it is important that it be durable and able to withstand chemical and mechanical forces during function. Water, enzymes, acids, and other chemicals present in saliva, pulpal fluid, and the demineralized dentin all may contribute to the degradation of this bonding layer.[16,22] This subject is discussed further later in this chapter. But it is essential that the adhesive layer be able to remain intact during the cyclic imposition of stresses that occur during chewing, clenching, and grinding. Studies of the durability of this bonded layer have been reviewed[23,24] and are ongoing. Most adhesives have been found to be at least somewhat susceptible to degradation over time, both in vitro[25] and in vivo,[26] the magnitude varying based on the formulation of the materials and the environmental conditions.

History of adhesion systems

Buonocore[27] first demonstrated the use of phosphoric acid to increase the receptivity of enamel surfaces to adhesives, producing a 100-fold increase in resin-enamel bond strength. Gwinnett and Matsui[28] subsequently found that fluid resin penetrated etched enamel prisms to form resin microtags that interlocked with superficial enamel irregularities (Fig 9-4). However, concerns at that time about the potential adverse effects on the pulp after contact with dentin limited the application of phosphoric

acid etching to enamel.

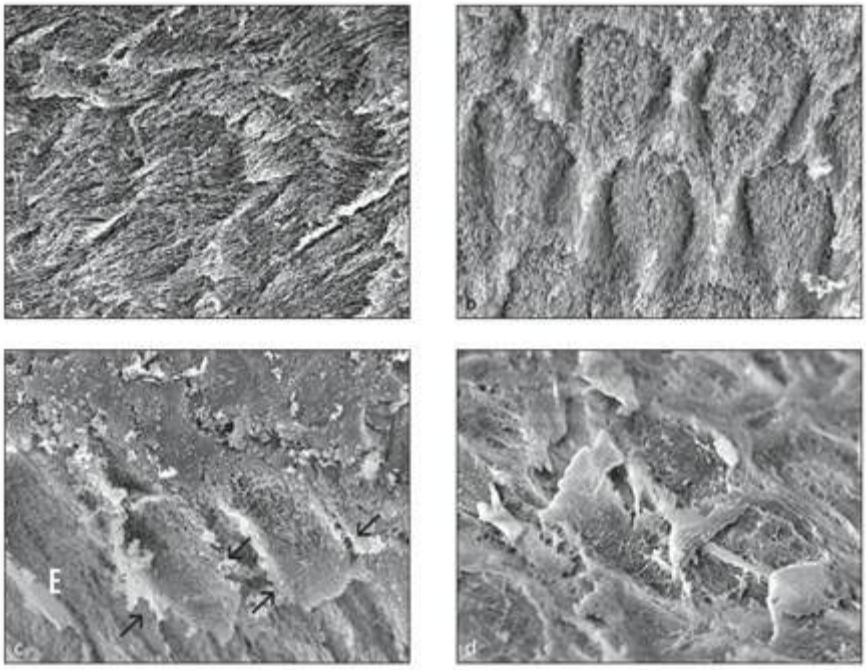


Fig 9-4 Adhesion to enamel occurs via penetration of liquid monomers that penetrate etched enamel prisms to form resin microtags that interlock with superficial enamel irregularities. *(a)* Field-emission scanning electron microscopy (FE-SEM) image of ground enamel showing minor irregularities produced by the underlying enamel prisms. *(b)* FE-SEM image showing the effect of 35% phosphoric acid on the enamel surface. Enamel prisms and hydroxyapatite crystals are clearly visible after acid etching. *(c)* FE-SEM image showing the interaction between an etch-and-rinse adhesive system and etched enamel (E). Resin tags *(between arrows)* penetrate the prisms' boundaries and in between the crystals, interlocking with the enamel surface. *(d)* FE-SEM image showing residual cured adhesive applied on phosphoric acid–etched enamel after complete removal of the enamel substrate. The irregularities in the adhesive surface are due to the interlocking between the liquid resin into the etched enamel surface before curing.

T h e *total-etching technique*, in which enamel and dentin are etched simultaneously, was introduced by Fusayama.[29] In this technique, the smear layer (Figs 9-5a and 9-5c) created on the surface by cutting with the bur is removed, and 3 to 5 μm of superficial dentin is demineralized to expose the collagen network (Figs 9-5b and 9-5d) and produce funnel-shaped dentinal tubules.[32]

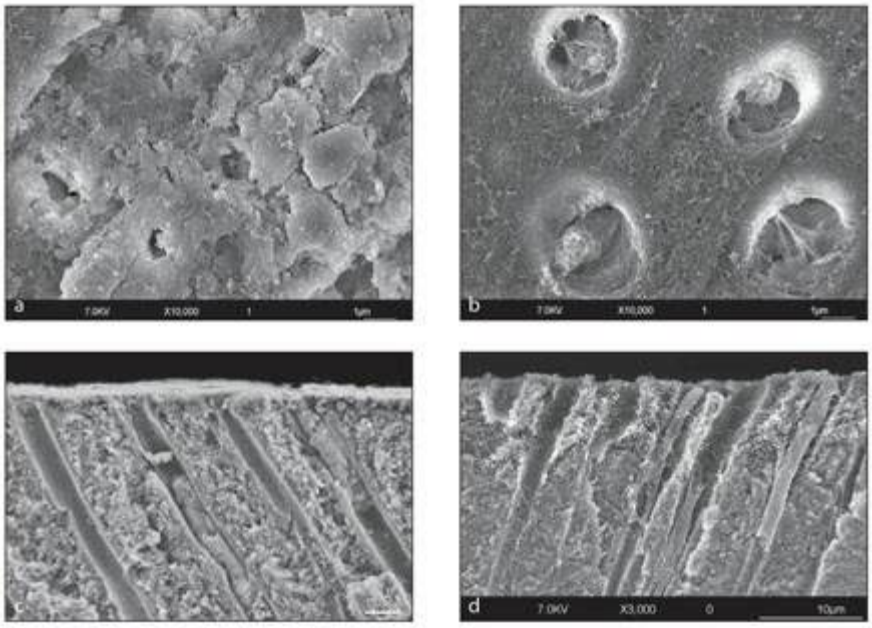


Fig 9-5*(a)* FE-SEM image of the smear layer obtained using a diamond bur with water irrigation on the dentin surface. An amorphous layer (approximately 1- to 5-µm thick) comprised of residual organic and inorganic debris as a result of mechanical instrumentation can be seen. This created smear layer obstructs the entrances of dentinal tubules, decreasing dentin permeability. *(b)* FE-SEM image showing the effect of 35% phosphoric acid on the dentin surface. The widened orifice of the tubule reveals the presence of intratubular structures suspended by a complex net of radial fibers connected to the surrounding inner lateral walls. The intratubular dentin appears as a porous structure characterized by a complex network of interconnecting collagen fibrills. (Reprinted from Breschi et al[30] with permission.) *(c)* Dentin cross section showing the smear layer on top of the mineralized dentin and smear plugs obliterating the tubules. *(d)* Etched dentin fractured longitudinally. Dentin tubule orifices appear funneled due to the demineralization of the peritubular dentin mineral. The intertubular dentin appears porous and demineralized up to 4 to 5 µm below the surface. (Reprinted from Breschi et al[31] with permission.)

Nakabayashi et al[33] first described the infiltration of resin monomers into acid-etched dentin (Figs 9-6a and 9-6b) to form a collagen fibril–reinforced resin matrix. This new structure was named the *hybrid layer* and is composed of collagen, resin monomers, and residual smear layer components (Figs 9-6c and 9-6d).

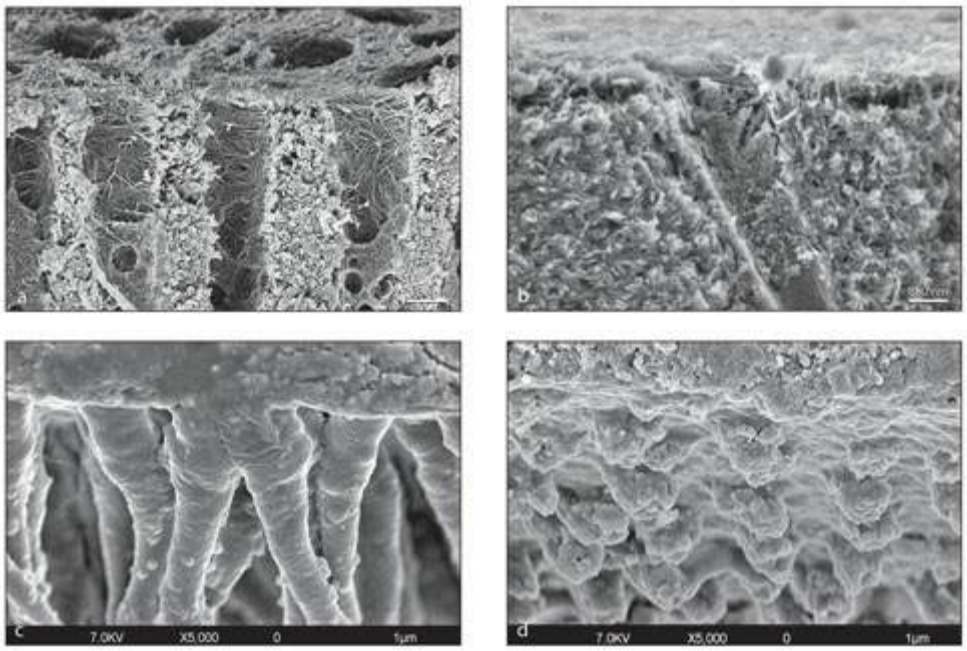


Fig 9-6 The hybrid layer formation occurs with penetration of the bonding systems into the acid-etched dentin (etch-and-rinse systems) or through a simultaneous demineralization and infiltration process (typical of self-etching systems). *(a)* FE-SEM image showing longitudinal view of dentin decalcification after etching with 35% phosphoric acid for 15 seconds (original magnification ×7,000). *(b)* FE-SEM image showing longitudinal view of decalcification of dentin after application of the etchant and primer agent of a strong self-etching adhesive (original magnification ×25,000). (Figs 9-6a and 9-6b courtesy of Sillas Duarte, Los Angeles, California.) *(c)* In-lens FE-SEM (FEI-SEM) image showing hybrid layer formation (continuous upper layer seen in the image) after the application of an etch-and-rinse adhesive. The image also shows the deep penetration of resin tags into the dentinal tubules exposed by the acid-etching process (image produced by demineralization of the dentin mineral to visualize the resin penetration). *(d)* FEI-SEM image showing hybrid layer formation (continuous upper layer seen in the image) after the application of a strong self-etching adhesive. The image also shows the minimal penetration of resin tags into the mildly etched dentin surface (image produced by demineralization of the dentin mineral to visualize the resin penetration). (Figs 9-6c and 9-6d reprinted from Breschi et al[22] with permission.)

The ability of dental adhesive systems to bond to tooth structures is currently recognized to depend on two main factors: *(1)* substrate demineralization, which partially removes the mineral phase and increases tooth receptivity, and *(2)* subsequent passive infiltration of monomers into the demineralized layer.[34,35] Cured adhesive forms a micromechanical bond with surface microporosities created during the demineralization process. Good bonding under clinical conditions depends on such interlocking and may also be influenced by other chemical interactions between functional monomers and the tooth substrate.[36,37]

Adhesive System Classification

Adhesives have been classified using several methods: generation, solvent type, mechanism of smear layer removal, and number of clinical steps.

Generation

Adhesives are grouped by product characteristics and order of market introduction, with each generation (first through eighth) reflecting significant changes in product characteristics.

Solvent

Adhesives are classified by solvent type and concentration. Commercially available adhesives have solvent concentrations of 8% to 49% and may be acetone, ethanol, or water based. The solvent contained in a given adhesive system has important clinical implications, particularly during air drying.[38] Many etch-and-rinse systems contain volatile solvents (eg, ethanol or acetone) that decrease viscosity and increase wetting and molecular mobility. Self-etching adhesives often contain water, which achieves the same effects and allows the dissociation of weak acids for enamel/dentin demineralization.[39] Organic solvents characterized by high vapor pressure (ie, evaporate easily) may contribute to the displacement of water from the collagen network and dentinal surface, allowing monomers to penetrate the collagen fiber network.[40,41]

Mechanism of smear layer removal

Adhesives are categorized by their approach to the removal of the *smear layer* (see Figs 9-5a and 9-5c). This 1- to 5-µm-thick amorphous layer is comprised of residual organic and inorganic debris and forms as a result of mechanical instrumentation of enamel and dentin.[42] The smear layer obstructs the entrances of dentinal tubules, decreasing dentin permeability by up to 86%.[43] This layer must be removed or made permeable to allow interaction between the monomers and the dentinal surface.

Based on the approach to smear layer removal, dental adhesives can be grouped into two major types[22,44]: *(1)* etch-and-rinse adhesives and *(2)* self-etching or etch-and-dry adhesives.

Etch-and-rinse adhesives

These adhesives use a strong acid (usually 35% to 37% phosphoric acid at a pH of

approximately 0.9) to completely etch enamel and dentin, followed by a water rinse to remove the acid from the tooth surface.[34] They formerly were known as *total-etch* adhesives. The acid removes the smear layer from the enamel surface and demineralizes the superficial hydroxyapatite to reveal the enamel prisms (see Fig 9-4). On dentin, the acid demineralizes the superficial hydroxyapatite and removes the smear layer and *smear plugs* (debris occluding the dentinal tubules) to expose the collagen fibrils of the dentinal matrix and open the dentinal tubules, funneling their orifices[34] (see Figs 9-5 and 9-6). The adhesive then fills enamel and dentin porosities created by the etching procedure (see Figs 9-4c, 9-4d, and 9-6c).

Self-etching or etch-and-dry adhesives

These systems use a nonrinsing solution of acidic monomers to dissolve the smear layer on enamel and dentin surfaces[35] (Fig 9-7).Because the self-etching or primer agent is simply air dried, they are also called *etch-and-dry adhesives*.[22] These adhesives render the smear layer permeable to monomers rather than removing it completely[35,44] (see Figs 9-7b and 9-7c). Some self-etching adhesives simultaneously dissolve the smear layer and infiltrate enamel and dentin,[45] using the mineral content of the substrate to buffer the acidic monomers and inhibit their dentin-etching ability with increasing depth.[46]

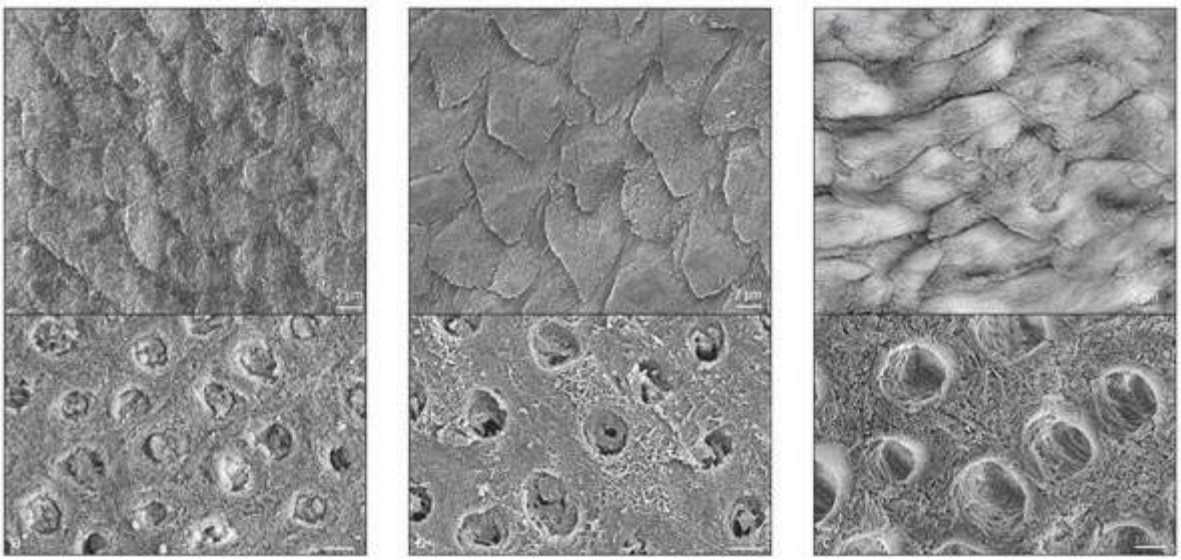


Fig 9-7*(a) (top)* SEM image showing mild decalcification on cut enamel due to the application of an ultramild self-etching adhesive (original magnification ×5,000). *(bottom)* FE-SEM image showing decalcification of dentin after application of ultramild self-etching adhesive (original magnification ×5,000). Note that some intertubular collagen is exposed, but smear layer remains within the tubules. *(b) (top)* SEM image showing decalcification patterns on cut enamel due to the application of a mild self-etching adhesive (original magnification ×5,000). *(bottom)* FE-SEM image showing decalcification of dentin after application of mild self-etching adhesive (original

magnification ×5,000). *(c) (top)* SEM image showing enamel etching patterns due to the application of a strong self-etching adhesive (original magnification ×5,000). *(bottom)* FE-SEM image showing decalcification of dentin after application of strong self-etching adhesive (original magnification ×5,000). Intertubular collagen is highly exposed, with no clear evidence of smear layer within the tubules. (Courtesy of Sillas Duarte, Los Angeles, California.)

Monomers with one or more attached carboxylic or phosphate-acid groups are considered to be self-etching. These adhesives are classified by strength according to chemical composition and acidity, which determine their morphologic features.[35,44] Ultramild self-etching adhesives (pH > 2.5) are able to demineralize a few hundred nm; mild adhesives (pH . 2) have about a 1-µm interaction depth; intermediate-strength adhesives (pH = 1 to 2) have an interaction depth of 1 to 2 µm, and strong self-etching adhesives (pH < 1) have an interaction depth of several µm.[35] The formation of resin tags in dentin tubules can be assured only with the use of strong self-etching adhesives; mild and ultramild adhesives may fail to form such tags, only slightly demineralizing smear plugs and allowing limited resin infiltration[35] (see Figs 9-6d, 9-7b, and 9-7c).

Number of clinical steps

In the most commonly used classification method, adhesives are grouped according to the number of clinical steps involved in the adhesion procedure.[44]

Etch-and-rinse adhesives may be three- or two-step systems: in *three-step systems*, separate etchant, primer, and bonding resin are applied consecutively; in *two-step (self-priming) systems*, etching is followed by the application of a combined primer and bonding resin.

Self-etching adhesives may be two- or one-step systems: in *two-step self-etching adhesive systems*, a combined etching and primer agent is applied on enamel and dentin and air dried, followed by the application and polymerization of a bonding resin, while *one-step self-etching adhesives* combine etching, primer, and bonding resin in a single application. Many one-step systems are not all-in-one solutions and require mixing materials from two or more bottles before application. Nevertheless, they are applied as one-step agents on the enamel or dentin substrate.

The two-step etch-and-rinse and one-step self-etching adhesives are considered simplified adhesives compared with their multi-step counterparts (three-step etch-and-rinse and two-step self-etching adhesives, respectively).[47]

Three-Step Etch-and-Rinse Adhesives

Three-step etch-and-rinse adhesives are fourth-generation products that require the sequential application of the three agents (Table 9-2 and Box 9-1). These agents are generally considered the gold standard against which other systems are compared. The procedure is described in detail and serves as a foundation for relating the function of each component in the adhesive systems. (See Fig 9-8 for step-by-step application of etch-and-rinse adhesives.)

Table 9-2	**Components, compositions, and application procedures of the common adhesives**		
Adhesive	**Manufacturer**	**Composition**	**Application procedure**
Three-step etch-and-rinse adhesives			
Adper Scotchbond Multi-Purpose	3M ESPE	Etching: 35% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 15 s. Remove excess water with an air syringe or by blotting. Leave moist.
		Primer: HEMA, polyalkenoic acid polymer, water	Apply to enamel and dentin. Dry gently for 5 s.
		Bonding: bis-GMA, HEMA, tertiary amines, photoinitiator	Apply to enamel and dentin. Light cure for 10 s.
All-Bond 3	Bisco	Etching: 32% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 15 s. Dry gently.
		Primer: Part A—ethanol, NGT-GMA; Part	Mix parts A and B (1:1) for 5 s. Apply 1 to 2 coats. Dry gently until

		B—bis-GMA, HEMA, BPDM	there is no visible movement of the material.
		Bonding: bis-GMA, UDMA, TEGDMA, glass filler	Apply one thin coat. Air dry/air thin. Light cure for 10 s.
Gluma Solid Bond	Heraeus Kulzer	Etching: 20% to 35% H_3PO_4	Etch 30 s on enamel. Etch 15 s on dentin. Rinse for 15 s. Dry gently.
		Primer: TEGDMA, HEMA, modified poly acrylic acid, maleic acid, ethanol, water, photoinitiators, stabilizers	Apply for 30 s. Dry for 1 to 2 s.
		Bonding: bis-GMA, TEGDMA, glass filler, SiO_2, photoinitiators, stabilizers	Apply and dry gently. Light cure for 40 s.
Optibond FL	Kerr	Etching: 37.5% H_3PO_4	Etch 15 s on enamel and dentin. Rinse for 15 s. Dry 3 s; do not desiccate.
		Primer: HEMA, GPDM, MMEP, PAMM, water, ethanol, CQ, BHT	Apply with light brushing motion for 15 s. Dry for 5 s.
		Bonding: bis-GMA, HEMA, GDMA, CQ, EDMAB, filler (fumed SiO2, barium, aluminoborosilicate, NaSiF6), coupling factor A174	Apply with light brushing motion for 15 s. Dry for 3 s. Light cure for 20 s.

Syntac	Ivoclar Vivadent	Etching: 37% H_3PO_4	Etch for 10 to 15 s on enamel and dentin. Rinse with water. Dry. Leave the dentin surface with a slightly glossy, wet appearance.
		Primer: TEGDMA, PEGDMA, maleic acid, dimethylketon, water	Apply; leave for 15 s. Dry thoroughly.
		Bonding: PEGDMA, glutaraldehyde, water	Apply for 10 s. Dry thoroughly. Light cure for 10 s.
Two-step etch-and-rinse adhesives			
Adper Scotchbond 1 XT Adhesive(also Adper Single Bond)	3M ESPE	Etching: 35% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 15 s. Leave tooth moist.
		Bonding: dimethacrylates, HEMA, poly alkenoid acid copolymer, 5-nm silane treated colloidal silica, ethanol, water, photoinitiator	Apply two consecutive coats. Dry gently for 5 s. Light cure for 10 s.
Excite F	Ivoclar Vivadent	Etching: 37% H_3PO_4	Etch for 20 to 30 s on enamel. Etch for 10 to 15 s on dentin. Rinse for at least 5 s. Leave tooth moist.
		Bonding: HEMA, phosphonic acid	Apply for at least 10 s.

		acrylate, dimethacrylates, silicone dioxide, initia tors, potassium fluoride in alcohol solution, stabilizers	Disperse the adhesive with a weak stream of air. Light cure for 10 s.
Gluma Comfort Bond	Heraeus Kulzer	Etching: 20% to 35% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 15 s. Dry for 1 to 2 s.
		Bonding: UDMA, HEMA, 4-META, modified polyacrylic acid, ethanol, glutar dehyde, water, photoinitiators, stabilizers	Apply three consecutive coats for 15 s. Dry gently. Light cure for 20 s.
One-Step Plus	Bisco	Etching: 32% H_3PO_4, 37% H_3PO_4 (Etch-37)	Etch for 15 s on enamel and dentin. Rinse thoroughly for 1 to 2 s.
		Bonding: BPDM, bis-GMA, HEMA, ace tone, photoinitiators, 8.5% by weight fluor aminosilicate glass fillers (1 μm)	Apply two coats to dentin. Agitate slightly for 10 to 15 s. Dry gently. Light cure for 10 s.
Optibond Solo Plus	Kerr	Etching: 37.5% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 15 s. Dry gently; do not desiccate.
		Bonding: bis-GMA, HEMA, GDMA, etha nol, CQ, EDMAB, BHT, filler (fumed SiO_2, barium aluminoborosilicate,	Apply with light brushing motion for 15 s. Dry for 3 s. Light cure for 5 to 10 s.

		Na_2SiF_6) coupling factor A174	
		Etching: 35% H_3PO_4	Etch for 15 s on enamel and dentin. Rinse for 5 s. Dry briefly; do not desiccate.
PQ1	Ultradent	Bonding: ethyl alcohol, methacrylic acid, 2-hydroxy methacrylate, 40% filler	Apply for 10 s. Dry from 10 mm for 10 s. Light cure for 20 s.
Prime & Bond NT	Dentsply/ Caulk	Etching: 36% H_3PO_4	Etch for 30 s on enamel. Etch for 15 s on dentin. Rinse for 15 s. Dry gently.
		Bonding: PENTA, di- and trimethacrylate resins, cetylamine, hydrofluoride, acetone, nanofiller (amorphous silicon dioxide, 8 nm), resin R5-62-1, T-resin, D-resin, CQ	Apply for 20 s (may necessitate additional coats). Dry for 5 s. Light cure for 10 s.
XP BOND	Dentsply/ Caulk	Etching: 36% H_3PO_4	Etch for 30 s on enamel. Etch for 15 s on dentin. Rinse for 15 s. Dry gently.
		Bonding: PENTA, TCB, HEMA, TEGDMA, UDMA, tert-butanol, functionalized amor phous silica, ethyl-4-dimethylaminobenzo ate, CQ, stabilizer, t-butanolate	Apply. Leave the surface undisturbed for 20 s. Air dry for at least 5 s. Light cure for a minimum of 10 s.

Two-step self-etching/etch-and-dry adhesives			
AdheSE	Ivoclar Vivadent	Self-etching primer: acrylic ether phos phonic acid, bisacrylamide, water, CQ, stabilizers	Apply for 15 s. Disperse the excess with a strong stream of air.
		Bonding: bis-GMA, GDMA, HEMA, fumed silica, CQ, tertiary amine, stabilizers	Apply. Dry with a very weak stream of air. Light cure for 10 s.
Clearfil Protect Bond	Kuraray	Self-etching primer: MDPB, 10-MDP, HEMA, hydrophilic dimethacrylate, photoinitiator, water	Apply. Leave for 10 s. Dry gently.
		Bonding: 10-MDP, HEMA, bis-GMA, hydrophobic dimethacrylate, photoinitia tors, silanated colloidal silica	Apply bond. Dry gently. Light cure for 10 s.
Clearfil SE Bond (Clearfil Mega Bond in Japan)	Kuraray	Self-etching primer: 10-MDP, HEMA, photoinitiator, water	Apply for 20 s. Gently air blow.
		Bonding: 10-MDP, bis-GMA, HEMA, hydrophilic dimethacrylate, microfiller	Apply two consecutive coats. Dry from 2 cm. Light cure for 10 s.
Optibond XTR	Kerr	Self-etching primer: GPDM, hydrophilic co-monomers, water, ethanol, acetone	Apply for 20 s. Dry thin for 5 s.
		Bonding: Resin monomers, HEMA, inorganic fillers, ethanolr	Apply with light brushing motion for 15 s. Light cure for 10 s.

Peak SE	Ultradent	Self-etching primer: bis-GMA, HEMA, ethanol, methacrylic acid, water	Apply for 20 s. Dry for 3 s.
		Bonding: bis-GMA, HEMA, ethanol, methacrylic acid, silica filler	Apply for 10 s. Dry for 10 s. Light cure for 20 s.
One-stepself-etching/etch-and-dryadhesives			
All-Bond SE	Bisco	Part I: ethanol, sodium benzene, sulfinate Part II: HEMA, BPDM, GPDM, bis-GMA	Mix parts A and B (1:1) until uniformly pink. Apply 1 to 2 coats, 5 to 10 s for each coat. Dry gently from 5 cm for 5 s. Light cure for 10 s.
Clearfil S^3 Bond	Kuraray	10-MDP, bis-GMA, HEMA, phosphate methacrylates, BHT, ethanol, fluorides, CQ, siliciumdioxide nanoparticles	Apply; leave for 20 s. Dry for more than 5 s. Light cure for 10 s.
Easy Bond	3M ESPE	HEMA, bis-GMA, methacrylated phosphor ic esters, HDDMA, methacrylate functional ized polyalkenoic acid (Vitrebond copoly mer), finely dispersed bonded silica filler with 7-nm primary particle size, ethanol, water, initiators based on CQ, stabilizers	Scrub adhesive for 20 s on dentin. Air thin for 5 s. Light cure for 10 s.
	GC	4-MET, phosphoric ester-monomer, UDMA, TEGDMA,	Apply. Leave undisturbed for 5 to 10 s. Dry

G-Bond	America	acetone, water, stabi lizer, silica, filler, water, photoinitiator	under maximum air pressure for 5 s. Light cure 10 s.
I Bond	Heraeus Kulzer	UDMA, 4-META, glutaraldehyde, acetone, water, photoinitiators, stabilizers	Scrub adhesive for 20 s. Gently air thin. Light cure for 20 s.
One-Up Bond F	Tokuyama	Part A: MAC-10, photoinitiator, methacry loylalkylacid phosphate, multifunctional methacrylic monomers Part B: MMA, HEMA, water F-deliverable, microfiller (fluoroaluminosilicate glass), photointiators	Mix thoroughly until evenly pink. Use bonding within 90 s after mixing. Apply for at least 10 s. Light cure for 10 s.
Xeno V	Dentsply/ Caulk	Bifunctional acrylic amides, acrylamide alkylsulfonic acid, functionalized phosphor ic acid ester, acrylic acid, CQ, butylated benzenediol, water, tert-butanol	Apply uniformly. Dry gently for 20 s. Light cure for at least 20 s.

Box 9-1	**Clinical application steps, advantages, disadvantages, and common errors for three-step etch-and-rinse adhesives**
Etching	
1. Apply 35% to 37% phosphoric acid to enamel for 15 to 30 s and to dentin for 15 s. 2. Rinse the etched surface for 15 s with an air-water spray. 3. Gently air dry to remove excess moisture. 4. Apply 0.2% to 2% aqueous chlorhexidine solution. 5. Remove excess chlorhexidine solution with a moist cotton pellet.	
Priming	
1. Perform active application of primer for at least 30 s (gently agitated or rubbed onto the dentin surface with	

a small brush). 2. Apply water-free, acetone-based primers generously in multiple layers. 3. The primed surface should appear glossy after air drying; if it appears chalky, the primer must be reapplied. 4. Primers must be adequately air dried to evaporate all of the solvent; usually 30 to 40 s are needed for proper solvent evaporation.		
Bonding		
1. Generously apply the adhesive resin with a microbrush. 2. Use a bristle brush to thin and create a homogenous layer. 3. Cure for manufacturer recommended time, which is typically about 10 to 20 s.		
Advantages	**Disadvantages**	**Common clinical errors**
• Phosphoric acid etch provides the best bond to enamel. • Several research reports support their use on different substrates, including metals and porcelain. • The highest dentin bond strengths among all dentin adhesives. • Generally contain a dual-curing option for indirect restorations and bonded amalgams. • Can be used with chlorhexidine rewetting for bond preservation.	• Multiple bottles make their use more cumbersome. • Possibility of running out of one component before another. • Because primer and adhesive resin are dispensed into separate wells in the same plastic container, their sequential application may be reversed. • Thick adhesives may pool easily around preparation line angles and margins.	• Overetching dentin. • Suboptimal rinsing of the etching gel. • Overwet/overdry dentin surface. • Insufficient primer application/penetration. • Insufficient primer solvent evaporation. • Overthinning bonding component. • Suboptimal polymerization of the bonding component.

Step 1: Etching

Given the differences in the histologic characteristics of enamel and dentin, the etching considerations for these substrates differ.

Etching of enamel

In enamel, etching substantially enlarges the surface area available for bonding and almost doubles the original surface energy[48,49] (see Fig 9-4). Three morphologic patterns of enamel etching have been described.[30,48,49] Type I etching removes the core prism material while leaving the periphery intact; type II etching removes the peripheral material while leaving the prism core relatively unaffected; and type III etching takes a more random approach characterized by a combination of types I and II that results in morphologically unclassified patterns. These etch patterns are determined by the mineral content and morphologic characteristics of the enamel,

and the dentist has no control over the pattern resulting from acid etching.

Etching creates enamel microporosities within which the resin tag extensions of monomers interlock micromechani-cally. *Macrotags* form between enamel-prism peripheries, and *microtags* form in prism cores.[28,44] These tags are formed within a 550-µm-thick microporous layer by phosphoric acid etching[50,51] (see Fig 9-4).

Enamel etching is affected by the content and concentration of the acid-etching solution and by the application technique and timing. These factors have been investigated extensively.[52–55] Many etching solutions have been tested, including those containing phosphoric (10% to 50%), fluoridated phosphoric, pyruvic, citric, maleic, oxalic, tannic, ethylenediamine-tetraacetic, trichloracetic, and polyacrylic acids[56–59]; however, most of the adhesive systems use phosphoric acid. The physical state of the solution also affects etching; gels can be applied to the enamel surface in a more controlled manner than liquids and achieve wider and deeper enamel penetration.[60] Etching agent application methods have also been investigated to improve clinical success; gels may be applied using a continuous brushing technique to obtain more etched enamel substrate and thereby improve the marginal adaptation of resin composite restorations.[61,62] Extending etching times to 60 seconds has been shown to produce rougher enamel surfaces but no enhancement of bond strength.[63]

Enamel etching is also affected by the chemical composition of the enamel (eg, high fluoride concentration makes enamel more resistant to etching). Moreover, aprismatic enamel is also more resistant to etching.[64]

Etching of dentin

Fusayama's revolutionary total-etching technique[29] to treat dentin was initially resisted by dentists, who feared that adverse pulpal reactions would result from the use of 40% phosphoric acid to etch dentin. Subsequent studies found that such reactions occurred because of bacterial leakage rather than acid use and that acid etching could thus be safely used on dentin more than 0.5 mm thick, provided the dentin was sealed after the etching.[32]

The use of phosphoric acid to etch dentin removes the smear layer, demineralizes the first 3 to 5 µm of dentinal tissue, and creates funnel-shaped dentinal tubules. The latter effect is because of the presence of higher mineral content in peritubular dentin than in intratubular dentin.[30] Acid etching thus renders the dentinal structure porous, allowing the impregnation of adhesive components[22,44] (see Figs 9-5 and 9-6).

Several studies have investigated the effects of different etching times on bond

strength.[34] Although extended etching times have been found to increase substrate porosity, they do not necessarily increase bond strength. The determination of an appropriate etching time should consider *(1)* the ability of monomers to impregnate the substrate in relation to dentin demineralization and *(2)* the ability of collagen fibrils to maintain integrity when exposed to phosphoric acid for a given length of time.

Clinicians should seek to demineralize dentin only to the desired extent of infiltration; excessive demineralization may produce a weak zone comprised of suboptimally impregnated dentin at the base of the hybrid layer consisting of exposed collagen fibrils.[65–68] Etching should be limited to superficial dentin because the viscosity of primers and bonding agents allows only a few micrometers of impregnation (see Figs 9-5 and 9-6). Several studies have found that prolonged acid-conditioning times resulted in fracture within the demineralized dentin zone when specimens were stressed to failure.[69,70] These findings indicated that excessive acid conditioning caused deeper demineralization of intertubular and peritubular dentin, rendering them more susceptible to incomplete infiltration by resin monomers. The reduced infiltration decreased bond strength by creating a weak zone where failure was likely initiated. The results of these studies thus indicate that sound dentin should be etched for no longer than 15 seconds. For this reason, etchant is initially applied to the enamel (Figs 9-8a and 9-8b), followed by application to the dentin (Fig 9-8c), to ensure adequate etching of enamel without overetching dentin.

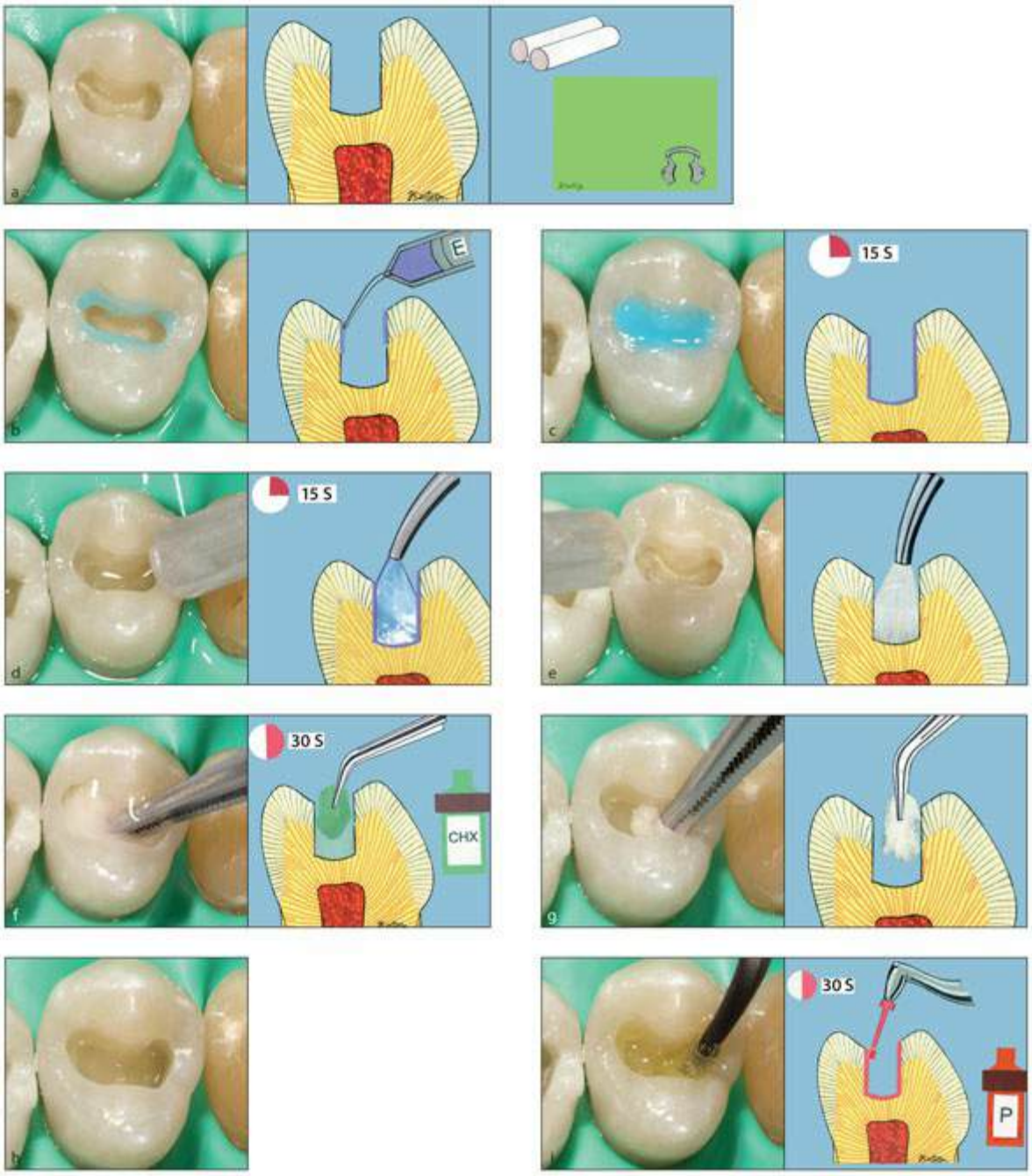


Fig 9-8 Step-by-step application of the three-step etch-and-rinse adhesives. *(a)* Preparation complete; ready for adhesive application. Appropriate isolation must be provided. *(b)* Phosphoric acid etchant gel is initially applied to the enamel margins only. *(c)* Phosphoric acid etchant gel is applied to the remaining prepared dentin surfaces and allowed to remain for 15 seconds. *(d)* Etchant gel is thoroughly rinsed with air-water spray for 15 seconds. *(e)* Preparation is dried. Frosty appearance of enamel confirms adequacy of etch. *(f)* The preparation is rewetted with an aqueous chlorhexidine solution for 30 seconds. *(g)* A damp cotton pellet is used to absorb excess chlorhexidine solution. *(h)* The glistening appearance of the dentin indicates the appropriate dentin moistness prior to applying the primer. *(i)* The primer component is actively applied for at least 30 seconds.

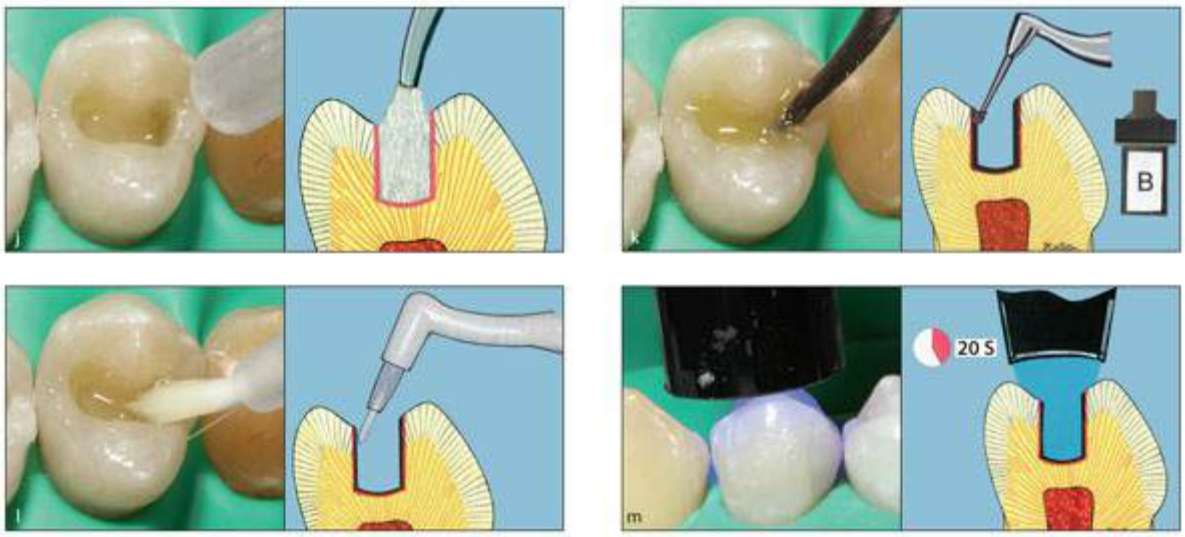


Fig 9-8*(cont) (j)* The primer component is air dried for 30 to 40 seconds to ensure thorough evaporation of solvents. *(k)* The adhesive component is actively applied to the entire preparation. *(l)* A bristle brush is used to absorb excess adhesive, avoid pooling, and distribute the adhesive evenly on all cavity walls. *(m)* The adhesive is light cured for 10 to 20 seconds prior to inserting the resin composite.

An immunohistochemical study of the collagen component of human dentin found that the aggressiveness of the etchant (pH and pKa) and the etching time affected the morphology of collagen fibrils and associated proteoglycans.[71] The exposure of dentinal collagen fibrils to an etching agent for more than 15 seconds may produce structural changes that weaken the hybrid layer.[72] However, in some cases, such as in the presence of aged or sclerotic dentin with a high mineral content, etching times of up to 30 seconds may be appropriate.[7,73]

After etching, the tooth should be rinsed with an intense air or water spray to remove the acid completely and stop the etching process (Fig 9-8d). The rinsing time should be similar to the etching time. When etching gels containing silica particles are used, particular care should be taken to rinse as long as instructed to ensure that all silica is removed from the dentinal tissue.[74]

After rinsing, the wetness of the dentin is an important clinical consideration. The first commercially available etch-and-rinse adhesive system was Clearfil Bond System-F (Kuraray), introduced in 1978. Clinicians who used this adhesive to etch dentin with 40% phosphoric acid followed by a dry-bonding technique reported adverse pulpal reactions.[34] In *dry bonding*, the cavity walls were dried after etching until the enamel margins were "frosty" to confirm the success of enamel etching. However, this technique has been found to cause the interfibrillar spaces between exposed dentinal collagen fibrils to collapse, preventing proper monomer

infiltration[75] (Fig 9-9). Although this technique achieved high resin-enamel bond strengths, resin-dentin bond strengths were insufficient to resist polymerization shrinkage forces. Thus, one or more walls would debond during resin composite polymerization, creating bacterial leakage through an increasingly permeable dentin that risked pulpal irritation.[32]

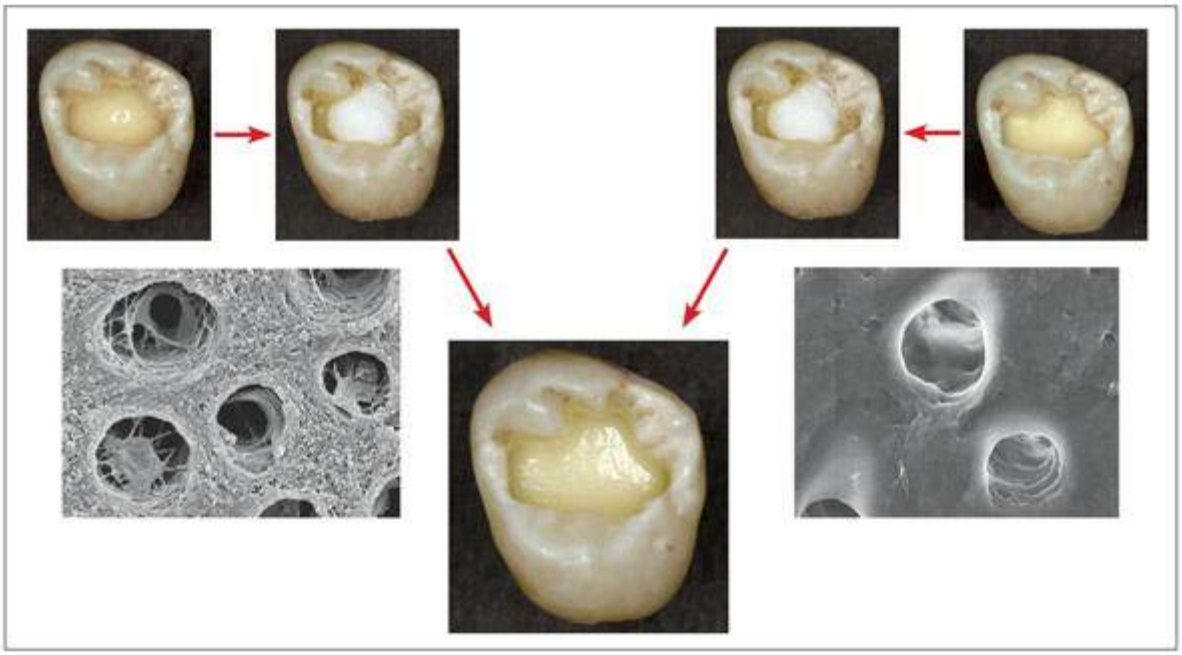


Fig 9-9 Wet bonding for the etch-and-rinse adhesive technique. The ultimate goal is to have a moist, glistening surface, as seen in the center photograph. If, after etching and rinsing, the surface is too wet, a moistened cotton pellet can be used to remove excess moisture (upper left). The SEM image on the left side shows large interfibrillar spaces in the intertubular dentin available for adhesive infiltration if the dentin surface remains moist. (Courtesy of Sillas Duarte, Los Angeles, California.) Conversely, if the dentin is overdried, a moist cotton pellet can be applied to the dried dentin for 30 seconds to remoisten the dentin surface (upper right). The SEM on the right side shows a compact intertubular surface due to collagen collapse. Under this condition, adhesive penetration is difficult to achieve due to the reduced intrinsic porosity of the collapsed collagen layer. (Reprinted from Breschi et al[30] with permission.)

The low resin-dentin bond strengths associated with dry bonding resulted in dentin sensitivity, microleakage, secondary caries, and loss of bonded restorations. To address these issues, Kanca[76,77] introduced the *wet bonding technique*, which used residual water on the dentin surface as a rewetting agent (see Fig 9-9). This technique increased resin-dentin bond strength, which provided a good dentin seal and reduced postoperative pain. By creating resin-dentin bonds that were equal or superior in strength to resin-enamel bonds, the wet bonding technique allowed clinicians to achieve safe, reproducible resin-dentin bonding.

Although the wet bonding technique[77] was proposed as a standard procedure with

the use of etch-and-rinse systems, it has not been standardized or made reproducible to date. Wet bonding can be performed using several different methods, and the proper degree of dentin wetness has not been determined.[78,79] The optimal surface wetness for wet bonding varies among acetone-, ethanol-, and water-based etch-and-rinse adhesive systems.[80] Additional factors, such as differences in the hydraulic conductance of superficial and deep dentin[81] and the presence of carious or sclerotic dentin in which whitlockite crystals completely or partially obliterate dentinal tubules,[82] prevent the achievement of uniform wetness on the axial, pulpal, and gingival walls. The same preparation may thus contain overly wet and overly dry regions, resulting in nonuniform resin bonding.

After the removal of pooled moisture by blotting or wiping with a slightly damp cotton pellet (see Fig 9-9), properly moist dentin should exhibit a shiny, hydrated surface.[83] Although drying techniques using air, absorbent paper, and cotton pellets have been proposed, absorbent moist cotton sponges[78,84] appear to be most effective. The rewetting of excessively dried dentin with water or a rewetting agent can raise collapsed collagen to a level comparable with that achieved with wet bonding.[80] Recently, the use of chlorhexidine has been proposed as a rewetting agent because of its capability to disinfect and stabilize the bond over time.[16,22,85–89] Bond stabilization occurs because chlorhexidine inhibits dentin endogenous enzymes that are claimed to participate in the bond degradation phenomena occurring at the adhesive interface[16,22] (see the sections on resin and collagen fibril degradation). Rewetting and bond stabilization can be accomplished simultaneously. After etching and rinsing, the cavity preparation is air dried (Fig 9-8e). In this way, the enamel can be assessed for adequate etching by checking for the frosty appearance of the enamel. The dentin can be rewetted with an aqueous chlorhexidine solution for 30 seconds (Fig 9-8f), and then a moist cotton pellet is used to absorb excess chlorhexidine solution and achieve the appropriate moisture level in the preparation (Figs 9-8g and 9-8h).

Step 2: Priming

Enamel priming

The application of a hydrophobic enamel bonding agent to air-dried enamel avoids the need for a separate primer application while still achieving effective bonding (see Fig 9-8e). Sealants can also be applied to acid-etched enamel without

additional treatment.[90] However, the enamel bonding process is not compromised by the application of primers to acid-etched enamel, and it may even be enhanced in situations where isolation is difficult. In wet bonding, primers should always be applied to acid-etched enamel to displace any residual surface moisture through solvent evaporation.

Dentin priming

Primers are mixtures of monomers, such as hydroxyethyl methacrylate (HEMA), triethyleneglycol dimethacrylate (TEGDMA), bis-GMA, and urethane dimethacrylate (UDMA), which have varying degrees of hydrophilic and hydrophobic properties (see Tables 9-1 and 9-2). Hydrophilic functionality facilitates monomer permeation into the collagen matrix to form a hybridized collagen-resin layer, and hydrophobic functionality facilitates restoration bonding to the resin matrix. Solvents are added to reduce the inherent viscosity of co-monomer blends, allowing them to infiltrate wet demineralized dentinal matrices[75,91] (see Table 9-2). The type of solvent in primer solutions has been found to significantly affect dentin bond strengths by influencing the ability to reexpand previously dried demineralized matrix.[92]

During rinsing of the etching agents, the presence of water maintains the full expansion of the demineralized dentinal matrix.[93,94] Subsequent air drying of dentinal surfaces removes most of the water from the matrix,[95] causing it to collapse in a manner similar to the collapse of collagen fibrils (see Fig 9-9). Collagen peptides come into contact with one another, forming new interpeptide hydrogen bonds that stabilize and stiffen the shrunken matrix.[96] Resin-dentin bonding is compromised by the lack of sufficient interfibrillar spaces available for resin penetration in such shrunken dentin.[75,97] To resolve these undesirable situations, the dentin must be rewetted and/or primers must be able to reexpand the collapsed matrix.[92]

Hybridization is believed to result from primer infiltration into the open network of the collagen matrix exposed by dentin demineralization and from subsequent in situ polymerization of the monomers present in the bonding agent. In wet bonding, some hydrogen-bound water remains in the intrafibrillar spaces of collagen as a water-bridged hydration network, even in completely demineralized dentin. Although solvents facilitate the replacement of water with adhesive monomers, primer monomer systems often fail to displace all intrafibrillar water.[98] This limitation suggests that monomer penetration into intrafibrillar spaces is somewhat restricted and that encapsulation and entanglement occur primarily in extrafibrillar

spaces.[98] After permeation, nonbonded, electrostatic, and hydrogen-bonding forces may thus develop between the monomer phase and collagen molecules.[99] Primers should be actively applied and agitated for at least 30 seconds on the dentin surface to improve penetration and enhance adhesion[100] (Fig 9-8i).

Primers should be gently air dried after application to volatilize any remaining solvent before the adhesive resin is applied (Fig 9-8j; see also Box 9-1). The extent of solvent evaporation depends on the clinician's technique and the co-monomer mixture.[101,102] Failure to adequately evaporate primer solvent will significantly adversely affect the bond to dentin and can have a more adverse affect on adhesion than just about any other application mistake.[103,104] Drying times of at least 10 seconds or longer are recommended.[17]

The effect of solvent type on the effectiveness of bonding and sealing of dentin has been the subject of numerous investigations. The results of one study suggested that primers with ethanol/water solvents show reduced technique sensitivity compared with those with acetone solvents, while verifying the importance of near-complete solvent removal for optimum results.[103]

Additional layers of primer should be added if the surface does not appear uniformly glossy.

Step 3: Bonding

Three-step adhesive systems use hydrophobic solvent-free monomer blends (usually bis-GMA and TEGDMA) as bonding agents (see Table 9-2). These agents should be applied to the primed surface with a brush and thinned to an optimal thickness of about 60 to 120 μm, depending on the viscosity of the adhesive (Figs 9-8k and 9-8l).

Because no solvent is present, using an air syringe to thin the adhesive resin layer should be used with care to avoid formation of a less homogenous surface, with some areas so thin that they do not cure[105,106] (ie, nothing but an oxygen-inhibited layer; see Fig 9-8l). Significant reductions in bond strength were observed when air thinning was compared with aggressive air thinning or brush thinning, suggesting that brush thinning may provide a better way to obtain a homogenous layer without over air thinning.[106] This is done by active application of the adhesive using a microbrush, then using a bristle brush to ensure an even coating of the resin (see Fig 9-8k; see also Box 9-1). Excess resin will be absorbed into the bristle brush. If the bristles become saturated, the brush can be squeezed in a gauze pad to remove the excess resin.

The adhesive resin should be properly cured before the restorative material is applied to ensure that successful adhesion is produced to the tooth surface (Fig 9-8m). The upper layer of the adhesive resin does not polymerize well because it is exposed to oxygen, which inhibits its cure, and for this reason is called the *oxygen-inhibited layer*. Therefore, the surface layer contains unreacted methacrylate groups that may be involved in copolymerization with the restorative resin.[107,108]

Adequate light intensity is an important factor in curing the resin layer; prolonged curing times that slightly exceed the manufacturer recommendations have been shown to improve polymerization and adhesive properties.[109,110] However, because of the potential adverse effect on the pulp because of temperature increase, especially with curing devices with high power,[111] manufacturer recommendations for curing time should be strictly followed.

Two-Step Etch-and-Rinse Adhesives

Two-step (simplified) etch-and-rinse adhesive systems were developed in an effort to simplify clinical application.[22,44,112] (See Fig 9-10 for step-by-step application of the two-step etch-and-rinse adhesives.) Considered as fifth generation, these systems employ an etching procedure similar to that for three-step systems, followed by the application of a combined primer and bonding resin (Box 9-2; see also Table 9-2). These simplified adhesives have been commercially successful because of their ease of use and reduced application steps.

Box 9-2	**Clinical application steps, advantages, disadvantages, and common errors for two-step etch-and-rinse adhesives**	
Etching		
1. Apply 35% to 37% phosphoric acid to enamel for 15 to 30 s and to dentin for 15 s. 2. Rinse the etched surface for 15 s with an air-water spray. 3. Gently air dry to remove excess moisture. 4. Apply 0.2% to 2% aqueous chlorhexidine solution. 5. Remove excess chlorhexidine solution with a moist cotton pellet.		
Priming and bonding		
1. Apply primer/bonding solution generously, producing a shiny appearance, then vigorously rub at least 30 s. 2. Air dry to evaporate solvent for 30 to 40 s. 3. Actively reapply the primer/bonding solution and air dry. 4. Cure for manufacturer recommended time, which is typically about 10 to 20 s.		
Advantages	**Disadvantages**	**Common clinical errors**
	• Most two-step adhesives	

• Phosphoric acid etch provides the best bond to enamel. • Laboratory research supports their use on enamel and dentin. • High immediate bond strength. • The combined primer/bond bottle concept makes them extremely user friendly. • Can be used with chlorhexidine for bond preservation.	showed lower bond strengths than their three-step counterparts(produced by the same manufacturer). • Acetone-based adhesives may lose their efficacy with constant utilization due to rapid evaporation of volatile components. • More coats than those recommended by the manufacturer often needed to maximize bond strength. • Thick adhesives may pool easily around preparation line angles and margins. • Some adhesives are not compatible with self-curing or dual-curing composites(core buildup composites and resin luting cements). • Inclusion of hydrophilic components in bonding resin can cause increased hydrolytic degradation.	• Overetching dentin. • Suboptimal rinsing of the etching gel. • Overwet/overdry dentin surface. • Reduced impregnation of the primer adhesive agent. • Inadequate solvent evaporation. • Overthinning adhesive when air drying to remove solvent; failure to use multiple coats. • Suboptimal primer/bonding polymerization.

Although two- and three-step etch-and-rinse adhesive systems produce resin-dentin bonds with similar strengths immediately following application,[44] long-term bond strengths differ between the two systems.[22,24,113] Two-step etch-and-rinse adhesives combine primer solvents and hydrophilic components with the more hydrophobic monomers of bonding agents (see Table 9-2), causing them to behave as semipermeable membranes.[114] The polymerization achieved with these adhesives is frequently suboptimal,[109,115] resulting in increased potential for water absorption from the underlying dentin.[116] These characteristics negatively impact long-term bond stability.

The combined primer and bonding resins also contain high concentrations (up to 50%) of solvents[91] (see Table 9-2) that may interfere with adhesive polymerization.[101] Although solvents decrease adhesive viscosity, facilitate co-monomer infiltration into acid-etched dentin, and increase the mobility of radicals and growing polymer chains,[98,101,117] solvent concentrations greater than 20% by weight usually lower the degree of conversion by increasing the physical space between reactive species during polymerization.[118]

The presence of residual water during dentin bonding may also adversely affect the curing reaction by preventing the formation of properly cured polymers within the hybrid layer.[119,120] This effect is likely because of the phase separation of hydrophobic and hydrophilic domains.[121]

Step 1: Etching

The considerations discussed above for etching with three-step systems also apply to the use of two-step etch-and-rinse adhesives (Figs 9-10a to 9-10d). The dentinal tissue moisture level is even more critical than with three-step etch-and-rinse adhesives because two-step etch-and-rinse adhesives do not have the rewetting ability of primers used in three-step adhesives. If the dentin is overdried, it should be rewetted prior to the next step (Figs 9-10e to 9-10h).

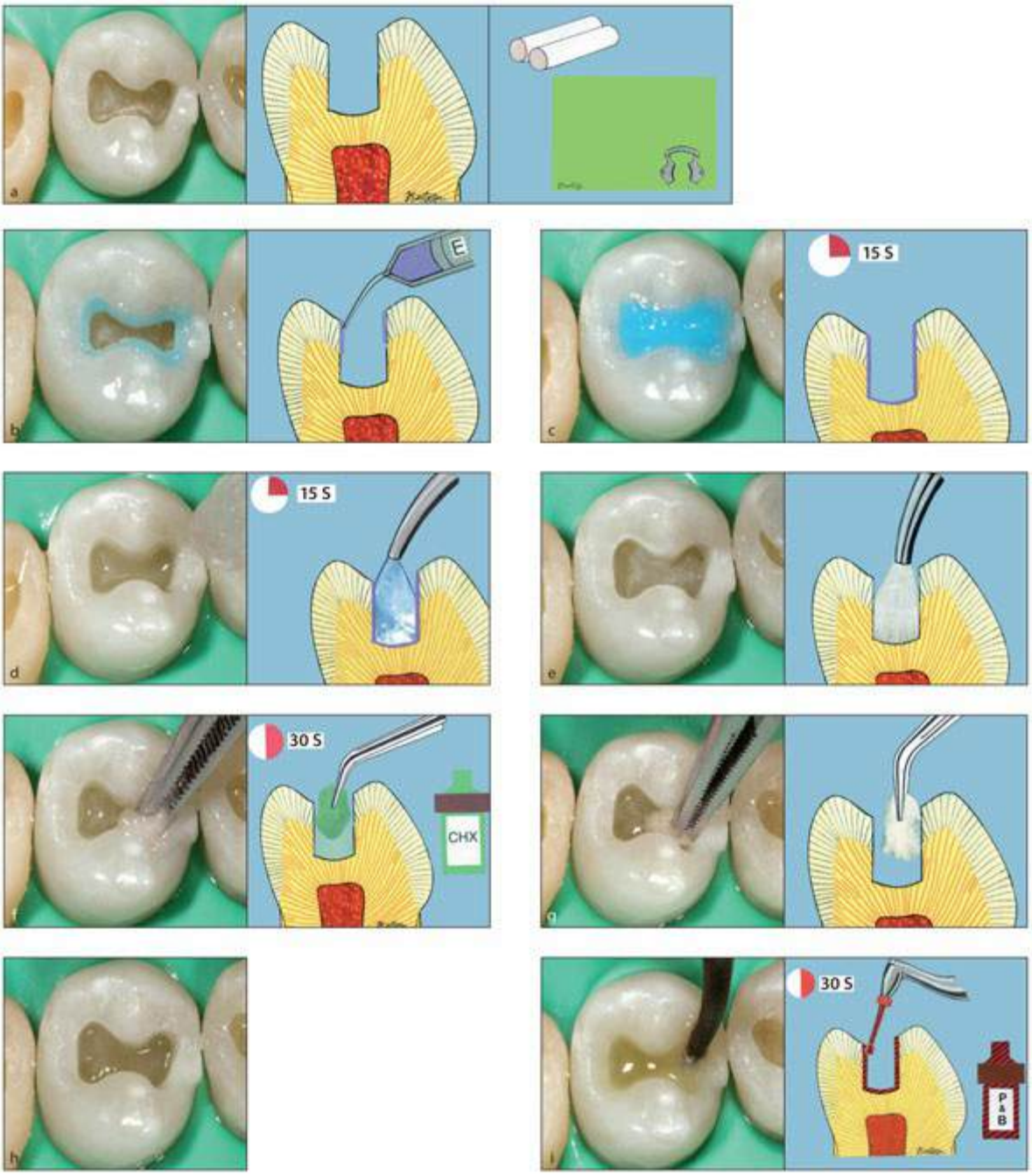


Fig 9-10 Step-by-step application of the two-step etch-and-rinse adhesives. *(a)* Preparation complete; ready for adhesive application. Appropriate isolation must be provided. *(b)* Phosphoric etchant gel is initially applied to the enamel margins only. *(c)* Phosphoric etchant gel is applied to the remaining prepared dentin surfaces and allowed to remain for 15 seconds. *(d)* Etchant gel is thoroughly rinsed with air-water spray for 15 seconds. *(e)* Preparation is dried. Frosty appearance of enamel confirms adequacy of etch. *(f)* The preparation is rewetted with an aqueous chlorhexidine solution for 30 seconds. *(g)* A damp cotton pellet is used to absorb excess chlorhexidine solution. *(h)* The glistening appearance of the dentin indicates the appropriate dentin moistness prior to applying the primer. *(i)* The combined primer-adhesive component is actively applied for 30 seconds.

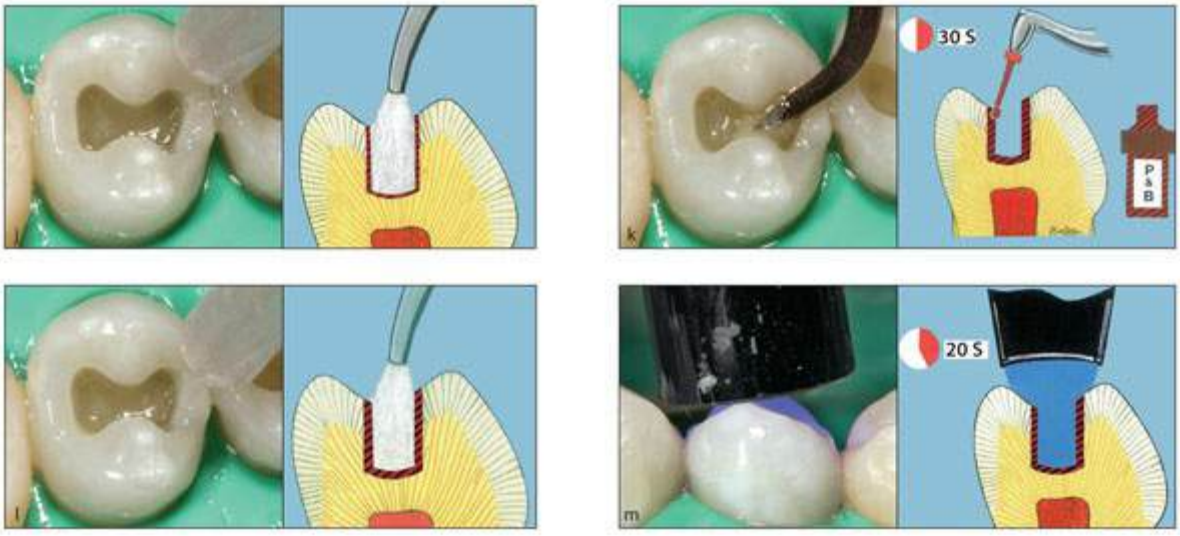


Fig 9-10*(cont) (j)* The primer-adhesive component is air dried for 30 to 40 seconds to ensure thorough evaporation of solvents. *(k)* The combined primer-adhesive component is actively reapplied for 30 seconds. *(l)* The primer-adhesive component is air dried for 30 to 40 seconds to ensure thorough evaporation of solvents. *(m)* The adhesive is light cured for 10 to 20 seconds prior to inserting the resin composite.

Step 2: Priming and bonding

Two-step etch-and-rinse adhesives contain a complex blend of monomers and solvents because of the combination of the amphiphilic properties of the primer and the more hydrophobic monomers of the bonding agent. Commercial adhesive co-monomer products commonly contain ethanol or acetone (see Table 9-2).

Several studies have found that the use of these adhesives results in the limited impregnation of demineralized dentin. Resin monomers have a limited capacity to diffuse into wet demineralized dentin,[122–124] producing a gradient of resin penetration with the highest concentration at the adhesive surface, lower concentrations within the hybrid layer, and little resin presence in the deepest portion of the demineralized zone.[125] Exposed and suboptimally impregnated collagen fibrils have been detected in the deepest areas of the hybrid layer with the use of these adhesives.[68]

Because of this difficulty impregnating the collagen fibril layer exposed by etching, the primer-adhesive agent should be applied with great care. Numerous clinical parameters have been found to affect the proper impregnation of the dentinal matrix (see Box 9-2):

Application time

Prolonged primer/bonding-agent application times increase monomer penetration into decalcified dentin and solvent/ water evaporation before light curing.[126,127] Recent studies have found that extended application times, even as little as 20 seconds, significantly increase the immediate and long-term bonding performance of the adhesives.[127–129]

Mode of application

The use of a rubbing action is essential to achieve a high immediate bond strength to dentin using acetone-, water-, and ethanol-based two-step etch-and-rinse adhesives[130] (Fig 9-10i). This action likely increases molecular kinetics and inward monomer diffusion in reaction to outward solvent diffusion, while also reducing dentinal wetness. Vigorous rubbing of demineralized dentin during the application of adhesive also improves the long-term stability of the dentin bond[131] by increasing the biomechanical characteristics of the hybrid layer.[132] Such vigorous application techniques can also improve the retention of restorations placed in noncarious cervical lesions, which typically pose a clinical challenge.[133]

Solvent evaporation

Before polymerization is performed, solvents should be completely evaporated to bring the reactant molecules into close proximity and to prevent residual solvent from plasticizing the polymer (Fig 9-10j). Adhesive solutions with higher solvent contents before light curing have lower degrees of conversion[101]; the higher solvent content also reduces the mechanical properties of adhesive polymers.[134,135]

When mixed with nonvolatile monomers, the evaporation capacity of solvents with relatively low vapor pressure (eg, water) is reduced as monomer concentration increases. It is thus impossible to achieve complete water evaporation from water-monomer mixtures.[136] The same principle applies to ethanol- and acetone-containing co-monomer mixtures; the nonvolatile monomer concentration increases as the solvent evaporates, thus reducing the vapor pressure of the remaining solvent and preventing complete solvent evaporation under clinical conditions. Thus, co-monomer composition as well as solvent type and content must be considered in determining optimal solvent concentration.

The adhesive layer must be carefully thinned, and application should be repeated to ensure adequate solvent evaporation with two-step etch-and-rinse adhesives[137] (Figs 9-10k and 9-10l). The use of a warm, dry airstream to evaporate the solvent

after the application of the primer/bonding agent may improve bond strength and hybrid layer quality by reducing the number of pores within the adhesive layer.[138]

Because adhesive solvents are volatile, the continuous transudation of dentinal fluid through open dentinal tubules before adhesive polymerization may create microscopic water-filled blisters along the adhesive interface.[116] After restoration placement, mastication may produce a pumping effect in these blisters that rapidly pushes fluid through the tubules, thereby potentially triggering the A-δ nerve fibers in the pulpal-dentin complex and causing sensitivity.[139]

Another issue is loss of solvent because of storage and handling. If the bottle is left uncapped, a significant amount of the volatile material can be lost rapidly. This results in alteration of the ratios from the manufacturer's intended formulation, which can then affect bonding. To avoid this, the bottle should be shaken and the material dispensed immediately prior to application.[140,141]

Curing time

Adhesive curing should be performed carefully to avoid compromised polymerization because of insufficient solvent removal, which results in the presence of high concentrations of hydrophilic monomers and water.[109,118] Manufacturer recommendations should be followed because resin permeability and monomer elution are related to the suboptimal polymerization of bonding resins[109,115,129] (Fig 9-10m).

Two-Step Self-Etching or Etch-and-Dry Adhesives

Because self-etching (etch-and-dry) adhesives contain acidic monomers that simultaneously condition and prime the dental substrate, they do not require a separate etching procedure (Boxes 9-3 and 9-4; see also Table 9-2). The salts removed from the enamel and dentin contribute to buffering of the acidity of the solution. Such systems have thus been described as more user friendly (shorter application time, fewer steps) and less technique sensitive (no wet bonding, simple drying) than etch-and-rinse systems.[35,44] Self-etching adhesives require only air drying, avoiding the need for rinsing after application.[22] Consequently, these adhesives do not remove the smear layer[35] (see Figs 9-5 to 9-7). While their use has been associated with lower postoperative sensitivity than that of etch-and-rinse adhesives, mainly when evaluated in deeper cavities[142,143] and likely because the

dentinal tubules remain partially obliterated by smear plugs,[144] other research has shown no difference between self-etching and etch-and-rinse adhesive systems in terms of postoperative sensitivity.[142,145–148]

Box 9-3	**Clinical application steps, advantages, disadvantages, and common errors for two-step self-etching or etch-and-dry adhesives**
Etching and priming	1. Selectively etch enamel with 35% to 37% phosphoric acid for 15 s. 2. Air-water rinse for 15 s, trying to minimize rinsing over the dentin. Dry gently. 3. Actively apply the self-etching primer agent on etched enamel and unetched smear layer–covered dentin for the time recommended by the manufacturer (typically 20 s). 4. Air dry to remove any excess solution and solvent and terminate the etching reaction.
Bonding	
1. Generously apply the adhesive resin with a microbrush. 2. Use a bristle brush to thin and create a homogenous layer.	

3. Cure for manufacturer recommended time, which is typically about 10 to 20 s.		
Advantages	**Disadvantages**	**Common clinical errors**
• No rinsing; quick application. No risk of overwet or overdry dentin. • Results of several studies support theiruse on dentin. • Additional chemical bonding is claimedto stabilize the hybrid layer for some adhesives. • Bonds well to enamel etched with phosphoric acid.	• If phosphoric acid etch not done first, enamel microleakage may result due to deficient enamel etch. • Prior etching of dentin surface with phosphoric acid may compromise bond to dentin. • Unknown effect of incorporating smear layer into adhesive zone. • Thick adhesives may pool easily around preparation line angles and margins.	• Insufficient etching on the enamel if selective preliminary enamel etch is not performed. • Inadvertent application of separate enamel etching agent to dentin.

Box 9-4	**Clinical application steps, advantage, disadvantages, and common errors for one-step self-etching or etch-and-dry adhesives**
	1. Selectively etch enamel with 35% to 37% phosphoric acid for 15 s. 2. Air-water rinse for 15 s, trying to minimize rinsing over the dentin. Dry gently. 3. Actively apply the one-step adhesive on etched enamel and unetched smear layer–covered dentin for the time recommended

Etching, priming, and bonding			by the manufacturer. 4. Air dry to remove any excess solution and solvent and terminate the etching reaction. 5. Reapply the adhesive in multiple layers using an active rubbing motion. 6. Air dry to remove any excess solution and solvent and terminate the etching reaction. 7. Cure for manufacturer recommended time, which is typically about 10 to 20 s.
Advantages	**Disadvantages**	**Common clinical errors**	
• Extremely simplified application procedure.	• Requires multiple layers. • Need for preliminary etching on enamel . • Lower bond strength than unsimplified counterparts (two-step systems). • Most of the adhesives are not compatible with self-curing or dual-curing composites (core buildup composites and resin luting cements). • Inclusion of hydrophilic components in bonding resin can cause increased	• Suboptimal polymerization. • Inadequate solvent evaporation. • Overthinning adhesive when air drying to remove solvent.	

	hydrolytic degradation.	

Two-step self-etching adhesives employ a self-etching primer followed by a hydrophobic and relatively solvent-free adhesive resin similar to that used in three-step etch-and-rinse adhesives (see Table 9-2 and Fig 9-5). These systems have demonstrated excellent performance under in vitro and in vivo conditions.[35,113,149] (See Fig 9-11 for step-by-step application of the two-step etch-and-rinse adhesives.)

Step 1: Self-etching and priming

Self-etching and priming of enamel

Clinically, the application of weak acids to enamel cannot produce the same extent of demineralization and typical frosty appearance obtained with the use of phosphoric acid[150,151] (see Fig 9-4). Self-etching primers produce a shallower and less retentive enamel-etching pattern because of insufficient penetration into the enamel surface.[35] In contrast to the macro- and microtags created by etch-and-rinse adhesives, self-etching primers intermingle primarily with the enamel surface to form *nanotags*. These tags are characterized by nanoretentive interlocking between enamel crystallites formed through inter- and intracrystallite resin penetration.[152]

Although some strong self-etching adhesives (pH < 1) create demineralized enamel that is morphologically similar to that created by phosphoric acid,[153] mild self-etching adhesives have shown unsatisfactory bonding to enamel, especially unground, aprismatic enamel.[35,154,155] The insufficient creation of enamel irregularities for penetration by self-etching adhesives results in low bond strength and poor marginal adaptation; thus, phosphoric acid should be preferred for enamel etching.[35,44,151]

The clinical use of self-etching adhesives on sound enamel is particularly problematic, especially in unbeveled preparations. Most studies[35,155–160] have confirmed that a preliminary separate etching procedure of enamel with phosphoric acid (> 15 seconds) should be included in the bonding procedure to achieve optimal enamel bonding results (Figs 9-11a to 9-11d; see Box 9-3).

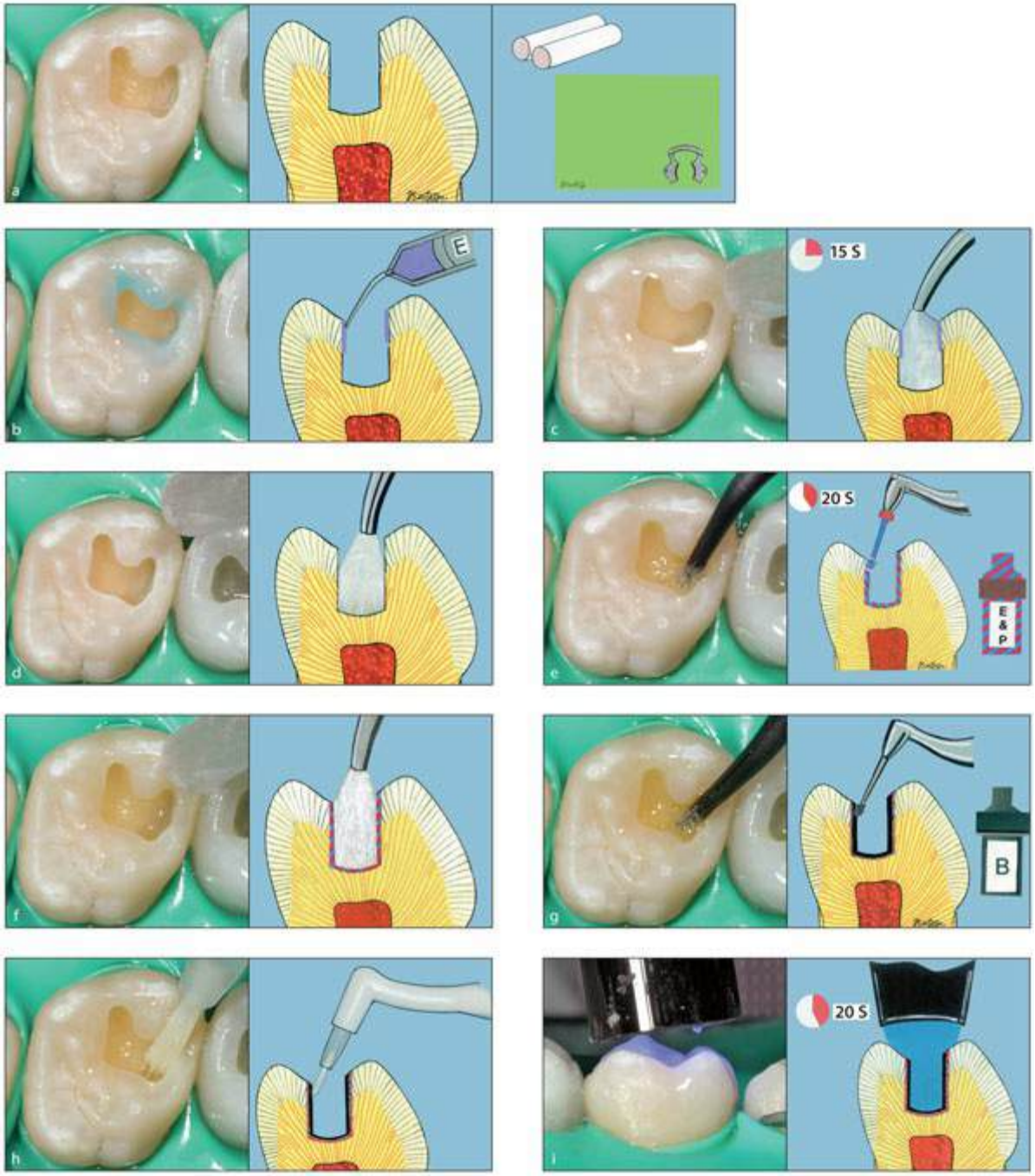


Fig 9-11 Step-by-step application of the two-step self-etching adhesives. *(a)* Preparation complete; ready for adhesive application. Appropriate isolation must be provided. *(b)* Phosphoric etchant gel is used to selectively etch the enamel margins of the preparation. *(c)* Etchant gel is thoroughly rinsed with air-water spray for 15 seconds. *(d)* Preparation is dried. Frosty appearance of enamel confirms adequacy of etch. *(e)* The combined etchant-primer component is actively applied for 20 seconds. *(f)* The etchant-primer-adhesive component is air dried for 30 to 40 seconds to terminate the etching reaction and ensure thorough evaporation of solvents. *(g)* The adhesive component is actively applied to the entire preparation. *(h)* A bristle brush is used to absorb excess adhesive, avoid pooling, and distribute the adhesive evenly on all cavity walls. *(i)* The adhesive is light cured for 10 to 20 seconds prior to inserting the resin composite.

Self-etching and priming of dentin

Although strong self-etching adhesives with improved enamel-etching performance are currently available, their ability to bond to dentin remains severely compromised. Their high intrinsic acidity does not allow complete buffering from the dissolved hydroxyapatite, leading to continuous etching[161] and incomplete polymerization.[115,161] Mild self-etching adhesives are thus preferred for dentin bonding.[35] Such adhesives partially demineralize dentin, leaving residual hydroxyapatite crystals that protect the collagen fibrils.[37]

Mild two-step self-etching adhesives have been found to exhibit excellent adhesion to smear layer–covered dentin, resulting in immediate bond effectiveness and longevity comparable with those achieved with ethanol-based, three-step etch-and-rinse systems that are sometimes considered the gold standard.[24,35,44,113,160] Two-step self-etching adhesives ensure simplified clinical application and reduced operator sensitivity by eliminating the need to etch dentin with phosphoric acid (see Box 9-3). The application of a self-etching adhesive to smear layer–covered dentin avoids most problems that are encountered when using etch-and-rinse adhesives, including excessive etching, suboptimal rinsing of the etching agent, and inappropriate adhesive application because of overwetting or overdrying of acid-etched dentin.[35,135,162] As mentioned previously for the etch-and-rinse systems, the primer-containing component should be actively applied to the preparation to enhance infiltration into the dentin surface and improve adhesion[127] (Figs 9-11e and 9-11f).

The reactive monomers contained in some mild two-step self-etching adhesives establish a chemical bond with the mineral component of dentin.[36,163,164] Specific functional monomers, such as 10-methacryloyloxydecyl dihydrogenphosphate (10-MDP), are responsible for this chemical interaction (see Table 9-2). Ionic bonds are formed between the carboxylic acid or phosphate groups within the functional monomers and the calcium of hydroxyapatite.[165] The resulting hybrid layer exhibits micromechanical interlocking (a prerequisite for good bonding) and allows additional chemical interaction between functional monomers and the tooth substrate through the adhesion-decalcification mechanism.[35]

The ionic interaction of functional monomers with the tooth may produce stable ionic bonds.[35,163,165] For this reason, the removal of the smear layer created by the demineralization of dentin during phosphoric acid etching depletes most of the hydroxyapatite in dentin that could be available for such chemical interaction, which in turn reduces the bond strength created by mild two-step self-etching adhesives.[166]

Thus, manufacturers of mild self-etching adhesive systems recommend that phosphoric acid etching be used only on enamel (see Fig 9-11b).

A major advantage of two-step self-etching systems is that they are relatively technique insensitive; thus, time or technique of application is not as critical for these systems as for other adhesives.[128,167]

A concern with the bonding process for self-etching adhesives is how to best maximize the bond to enamel without compromising the bond to dentin. As noted previously, a separate application of phosphoric acid to enamel prior to self-etching adhesive application will significantly improve the bond to enamel. However, there is considerable evidence that etching dentin with phosphoric acid prior to application of a self-etching adhesive significantly reduces bond strength to dentin, likely because of poor infiltration into demineralized dentin.[168–170] Because it is often difficult to apply phosphoric acid etchant to enamel without involving dentin, this procedure must be approached with caution. Clinical studies in which this approach was used did not demonstrate adverse clinical performance in Class 5 restorations over 2 years[171] and 5 years,[172] demonstrating that careful clinical application using a selective phosphoric acid etching technique with self-etching systems can be successfully employed with some restorations.

Step 2: Bonding

The bonding agents of two-step adhesives contain hydrophobic and relatively unsolvated monomer blends (usually bis-GMA and TEGDMA; see Table 9-2). Thus, the clinical recommendations described above for three-step etch-and-rinse adhesives are also applicable to two-step self-etching systems (Figs 9-11g and 9-11h; see Box 9-3). These include brush thinning to optimize adhesive thickness, the use of a layered application to achieve a fully saturated surface, and the use of optimal curing times (Fig 9-11i).

One-Step Self-Etching or Etch-and-Dry Adhesives

One-step self-etching adhesives are the simplest and most recently developed adhesive systems. (See Fig 9-12 for step-by-step application of the one-step adhesives.) These systems are classified as seventh- or eighth-generation adhesives because they may be multibottle (mixed just prior to use) or single-bottle (all-in-one) systems, respectively. One-step self-etching adhesives combine the three steps of the adhesion process into the application of a single solution containing complex

mixtures of hydrophilic and hydrophobic resin blends, acid, and water to activate etching (see Table 9-2). This chemical complexity has caused shelf-life problems.[173]

This class of adhesives has the highest hydrophilic monomer content of currently available adhesive systems; some of the earliest one-step adhesives are more acidic and hydrophilic than two-step self-etching primers[44,112] (see Table 9-2). Although these adhesives simplify clinical application, they result in lower immediate bond strengths than that produced with more complex adhesive systems.[35,44]

Adhesive interface stability may also be compromised by the use of one-step self-etching adhesives because of expedited aging phenomena affecting the hybrid layer.[22,24] The high hydrophilic monomer content of these formulations produces an adhesive interface after polymerization that acts as a semipermeable membrane on the dentin surface, allowing fluids to cross the interface because of osmotic pressure gradients.[154,174,175] Simplified adhesives thus demonstrate increased water sorption, which promotes polymer swelling and other water-mediated degradation phenomena.[15,22,24]

Although hydrophobic dimethacrylates have been added to one-step adhesives to produce stronger cross-linked polymer networks (see Table 9-2), the hydrophilic monomers tend to cluster before polymerization begins to create hydrophilic domains[121,176] and microscopic water-filled channels (water trees).[177] These channels permit the movement of water from the underlying dentin through the hybrid and adhesive layers to reach the adhesive-composite interface.[175,178]

One-step self-etching adhesives have also shown lower degrees of conversion than have multistep systems.[109,115] The high concentration of hydrophilic domains and the presence of water result in suboptimal polymerization and reduced bond longevity because of the elution of unreacted monomers. This process creates a porous structure with reduced sealing ability along the adhesive interface.

Application to enamel

Like two-step self-etching adhesives, one-step systems have demonstrated a reduced ability to bond to unabraded enamel.[35,154,158,159,179] This clinically relevant problem is particularly pronounced for unbeveled preparations. For this reason, preliminary phosphoric acid etching of enamel (especially non-instrumented enamel) is recommended before the application of one-step adhesives[159,166] (Figs 9-12a to 9-12c; see Box 9-4).

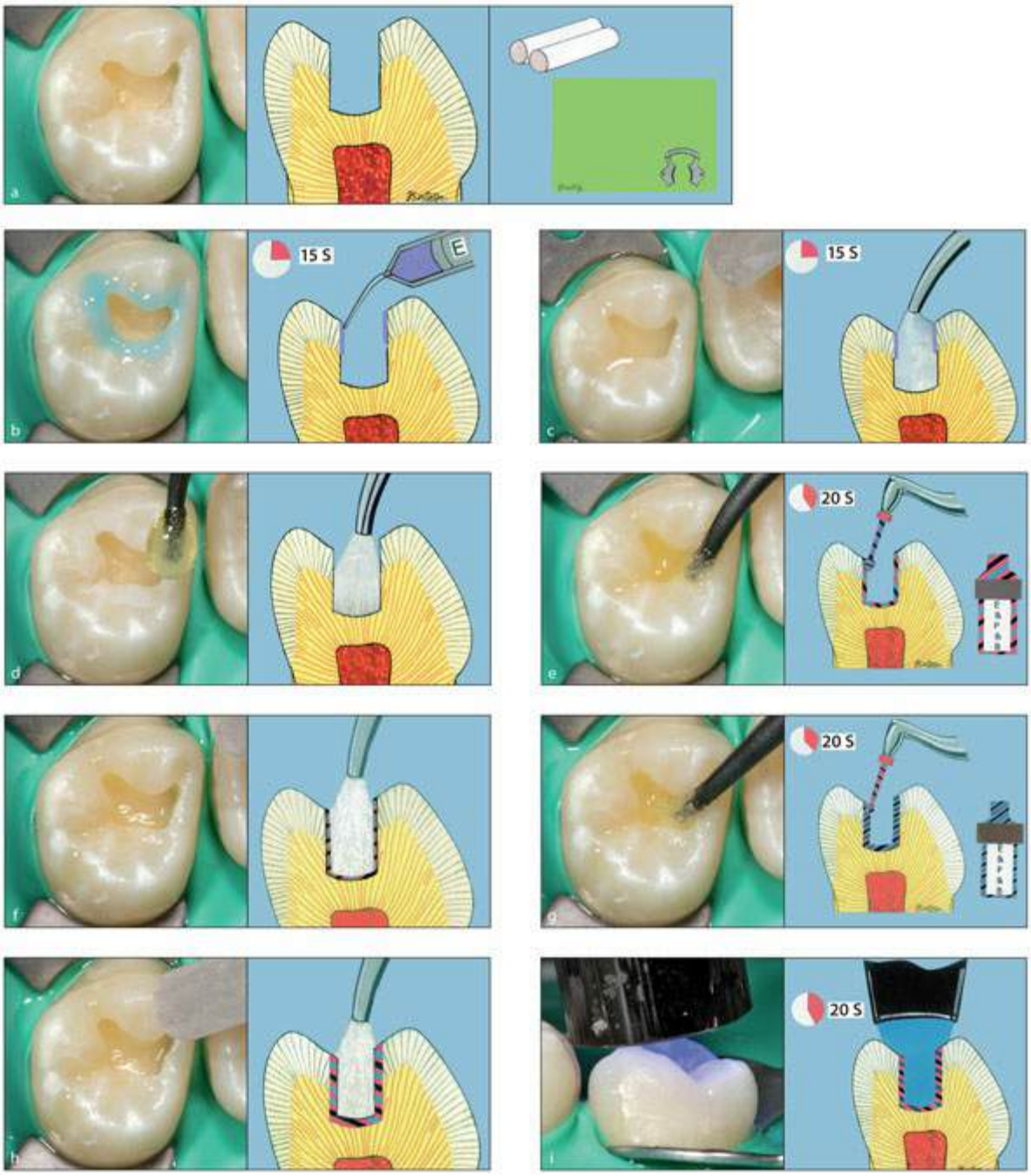


Fig 9-12 Step-by-step application of the one-step self-etching adhesives. *(a)* Preparation complete; ready for adhesive application. Appropriate isolation must be provided. *(b)* Phosphoric etchant gel is used to selectively etch the enamel margins of the preparation. *(c)* Etchant gel is thoroughly rinsed with air-water spray for 15 seconds and then air dried. *(d)* Preparation is dried. Frosty appearance of enamel confirms adequacy of etch. Applicator brush with combined etchant-primer-adhesive component is ready for application. *(e)* The combined etchant-primer-adhesive component is actively applied for the recommended time. *(f)* The etchant-primer-adhesive component is air dried for 30 to 40 seconds to terminate the etching reaction and ensure thorough evaporation of solvents. *(g)* The combined etchant-primer-adhesive component is actively reapplied. *(h)* The etchant-primer-adhesive component is air dried for 30 to 40 seconds to terminate the etching reaction and ensure thorough evaporation of solvents. *(i)* The adhesive is light cured for 10 to 20 seconds prior to inserting the resin composite.

Application to dentin

Efforts to improve the stability of the dentin-adhesive interface created by one-step adhesives have focused on several parameters: preliminary etching, hydrophobic coating, application mode, application times and multilayering, and appropriate polymerization.

Preliminary etching

As mentioned for the two-step self-etching systems, it can be clinically difficult to prevent the extension of phosphoric acid from enamel to dentin during etching, especially in small Class 2 preparations. Thus, preliminary phosphoric acid etching of dentin before the application of a self-etching adhesive has not been recommended (see Fig 9-12b and Box 9-4). Although preliminary dentin etching should be avoided with the use of adhesive formulations containing reactive monomers,[35,166] a recent study found that the application of some one-step adhesives to acid-etched dentin improved adhesion capability in comparison with application to smear layer–covered dentin.[159]

Self-etching adhesive products are intended to eliminate the need for phosphoric acid etching of dentin, thus simplifying clinical application and substantially reducing operator technique sensitivity. The effectiveness of an adhesive system applied to acid-etched dentin can be reduced dramatically by the potential operator-related problems mentioned above, including excessive etching, suboptimal rinsing of the etching agent, and inappropriate adhesive application because of overly wet or dry acid-etched dentin. Postoperative sensitivity may also result from increased dentin permeability after etching. To reduce operator sensitivity, the pre-etching of dentin before the application of one-step adhesives should be avoided.

Hydrophobic coating

One-step self-etching adhesives may be treated as primers that are subsequently covered with a hydrophobic coating, such as those employed in conventional three-step etch-and-rinse systems. This approach may improve adhesive performance; the application of an additional hydrophobic resin layer has been shown to improve immediate resin-dentin bond strength and reduce long-term adhesive interface degradation.[170,180–182]

Application mode

Active application produces consistent etching and enhances the interaction of acid monomers with the tooth structure, dispersing etching by-products into the surface (Figs 9-12d and 9-12e; see Box 9-4). The agitation of one-step adhesives brings fresh acidic solution into contact with the substrate, disperses air trapped in the solution, and mixes etching by-products to increase the efficiency of removal.[183] The use of an active brushing technique increases the immediate bond strength produced by one-step adhesives[184,185] and improves long-term stability.[183,184]

Application times and multilayering

The use of multilayering techniques or prolonged application times during dentin bonding[167,182,186,187] has been suggested to enhance the uniformity of adhesive infiltration and increase water and solvent evaporation[167] (Figs 9-12e to 9-12h; see Box 9-4).

Appropriate polymerization

The use of extended curing times that exceed the manufacturer recommendations has been found to improve polymerization and reduce permeability, potentially improving the performance of one-step adhesives[109,115] (Fig 9-12i; see Box 9-4).

Multimode or Universal Adhesive Systems

Dental manufacturers have recently made slight modifications of dentin adhesive formulations to produce a new class of universal adhesives. These materials are called *multimode* or *universal* because they can be used as self-etch, etch-and-rinse, or selective-etch systems. These adhesives have the ability to bond methacrylate-based restoratives, cement, and sealant materials to dentin, enamel, glass ionomer, and several indirect restorative substrates, including metals, alumina, zirconia, and other ceramics. All are modeled after self-etch systems and contain acidic monomers (see Table 9-1). Some earlier self-etch systems were noted to be negatively affected by pre-etching of dentin with phosphoric acid. Although these universal adhesives are new and just now being studied, early research appears to show that they are not adversely affected by this etch-and-rinse step either in vivo[188] or in vitro.[189,190]

The reported strong bond strengths of these new materials to metal and zirconia is likely a result of a functional acidic monomer MDP contained within the formulation

of most universal adhesives. This reactive monomer has been reported to have a strong affinity to dentin and enamel[191] and metal.[192] In addition, these materials form a chemical bond with zirconia (Zr-O-P).[193] It has also been shown that MDP-containing adhesives render the adhesive interface more resistant to biodegradation[194] because of the chemical bond with the residual hydroxyapatite crystals that remain available on the substrate after use of a self-etch approach on dentin. Some of these new formulations also contain silane coupling agents to aid adhesion to certain ceramic surfaces.[191]

The primary use of these adhesives is with light-activated resin composites in direct restorations. Nevertheless, because of the limited thickness of the adhesive layer, they can also be used to lute indirect restorations with self- or dual-cured composites and cements in combination with a self-curing activator. If the initial research on these materials is confirmed by longer-term studies, these adhesives should provide greater flexibility to the practitioner.

Role of the Smear Layer on Dental Adhesives

Cavity preparation is accomplished through the use of manual or mechanical instrumentation (ie, different burs or hand instruments) producing a smear layer both on enamel and dentin. Depending on the type of burs and their coarseness and the speed of the handpieces or the type of manual instruments, the thickness and composition of the smear layer produced may greatly vary.[195–198]

Medium-grit (100-μm) diamond bur preparation on enamel results in a rough and irregular surface with enamel fragments fractured or loosely attached to the enamel surface.[155] Moreover, numerous subsurface cracks are present. Dentin surfaces prepared with medium-grit diamond burs show more irregularities associated with a thicker (approximately 5-μm) and denser smear layer resulting in a compact smear plug that occludes dentinal tubules when compared with dentin prepared with tungsten carbide burs (smear layer thickness of approximately 2 μm).[199]

The smear layer is not strongly attached to the tooth substrate, and the adhesive systems should remove or modify it to permit an adequate impregnation of the underlying tooth surface and thus proper hybrid layer formation.[35] While etch-and-rinse adhesives completely dissolve the smear layer, self-etching (etch-and-dry) adhesives only modify it, and so their bond strength can be affected by the smear layer thickness.[35] Smear layer modification is correlated to the pH of the self-etching adhesives.[44] Strong self-etching adhesives are able to dissolve the smear layer,

while mild and ultramild self-etching adhesives cannot perform as well.[200] A thick smear layer can potentially impair the infiltration process of ultramild adhesives because the acidic monomers are neutralized by its components at an early stage, reducing the etching effect on the dentin.[196] It has been demonstrated that when ultramild self-etching adhesives are used, the thicker the smear layer, the lower the bond strength.[35,199,201] Thus, if a mild or ultramild adhesive will be used, an attempt should be made to reduce the smear layer thickness, such as by lightly abrading all cavity walls using fine-grit diamond burs.[202] Prolonged application time and active brushing of the adhesive during its application may also help to dissolve the smear layer, thereby promoting substrate infiltration and improving the bond strength of self-etching adhesives to dentin.[199]

Similarly, the smear layer is thought to interfere with the development of an adequate adhesive layer at the enamel interface when mild and ultramild self-etching adhesives are used. Because the smear layer is more easily dissolved than intact enamel, its neutralizing effect on acidic monomers of self-etching adhesives is faster. As a result, a thick smear layer might act as a barrier to adequate demineralization of the enamel substrate.[155] For this reason, the selective etching of enamel with phosphoric acid when using self-etching adhesives should always be performed.[35,155]

Glass-Ionomer Cements

Glass-ionomer cements, also called *polyalkenoates*, are the only dental materials in which the primary adhesion to tooth surfaces is via a true chemical bond.[203] Manufacturers provide these cements as two separate components: a powder that contains calcium aluminosilicate glass and a liquid that contains polyacrylic acid (and, in some cases, maleic acid) in water[203] (Fig 9-13). Setting occurs when the acid groups react with cations dissolved from the glass surface to form a salt.[204]

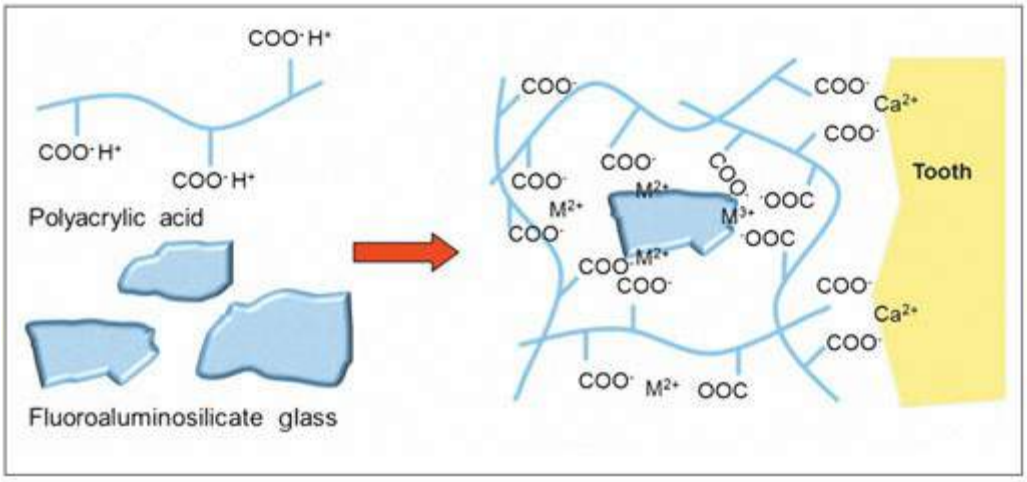


Fig 9-13 Glass-ionomer cement setting reaction. The carboxyl groups of the acids contained in the glass-ionomer material interact with the calcium content of tooth hydroxyapatite to form ionic bonds.

The carboxyl groups of the acids contained in the glass-ionomer material interact with the calcium content of tooth hydroxyapatite to form ionic bonds[204] (see Fig 9-13). When glass-ionomer cements are used to bond to dentin, smear layer removal and partial hydroxyapatite demineralization are frequently achieved through the application of a weak polyalkenoic acid conditioner. This procedure coats the collagen fibrils with hydroxyapatite and produces a pore-interspersed network.[205,206] The glass-ionomer material then acts as a self-etchant by hybridizing the partially demineralized dentin.[44] Some glass ionomers and polyalkenoic acid conditioners form a gel phase composed of calcium polycarboxylate salt at the bonded interface.[207]

The use of glass-ionomer cements produces bond-strength values similar to those of two-step self-etching adhesives on enamel and equal to those of one-step self-etching adhesives on dentin.[44] However, it is difficult to determine the true bond strength of glass ionomer to tooth structure, because examination of the failure surfaces typically shows residual glass-ionomer on the bonding surfaces, suggesting cohesive failure within the material.[208]

Adhesion to Intraradicular Dentin

Posts are routinely used to restore endodontically treated teeth because of the favorable properties of fiber posts in combination with several luting agents and adhesive systems.[209,210] However, the selection of a luting strategy that provides

reliable and long-lasting bonding to root dentin is difficult because of the large variety of products and the intrinsic difficulties of bonding within the endodontic space.

Posts can be cemented using conventional dual-curing resin-based cements in combination with etch-and-rinse or self-etching adhesives or using the recently formulated self-adhesive cements, which are all-in-one resin-based materials that allow simultaneous bonding between the intraradicular dentin and the post[211,212] (Figs 9-14 and 9-15). The procedure of bonding to intraradicular dentin differs from that to coronal dentin mainly because of the limited access to the endodontic or post space, which may lead to clinical errors that compromise adhesion.[213,214]

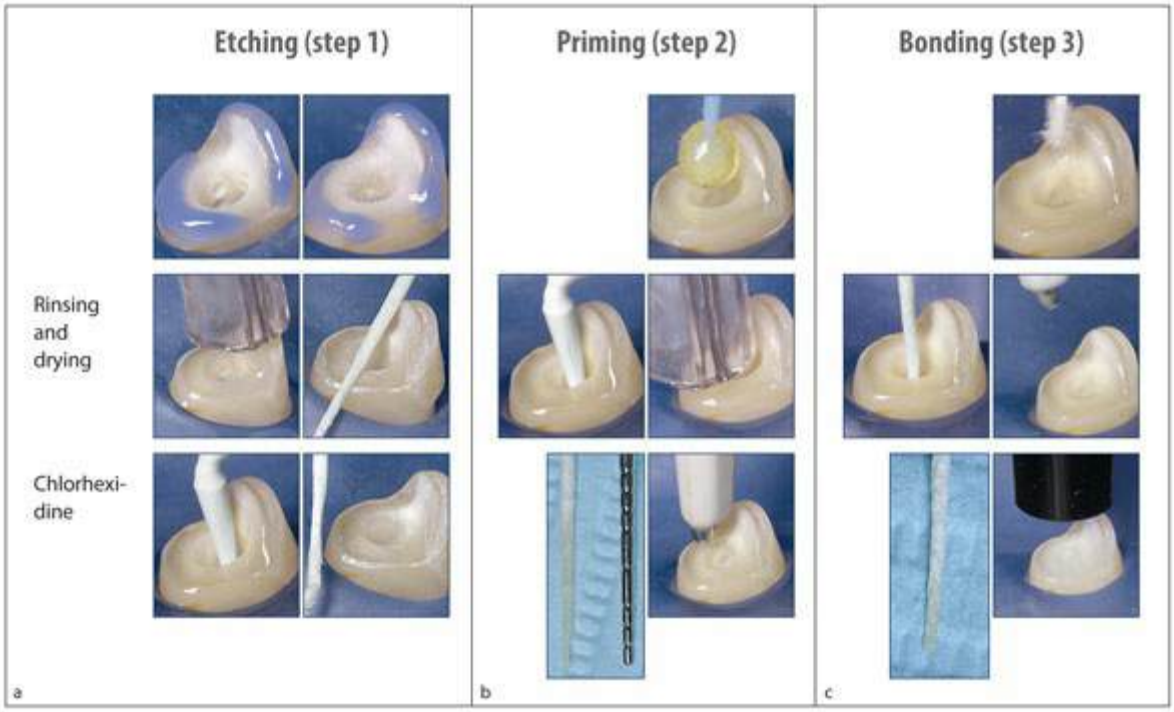


Fig 9-14 Step-by-step procedure of intraradicular bonding using a three-step etch-and-rinse adhesive. (Courtesy of Adamo Monari, Verona, Italy.) *(a)* In the etching step (1), enamel is etched with phosphoric acid gel, followed by placement of liquid phosophoric acid on the dentin and into the canal. This is to enhance diffusion of the acid down the canal and to improve rinsing for acid removal. Note that a paper point is used to ensure adequate removal of water after rinsing. *(b)* In the priming step (2), a thin microbrush is used to apply the primer, a paper point is used to remove excess primer, and it is gently air dried to evaporate the solvent. A periodontal probe is used to measure the extent of the prepared canal and is compared to the saturated length of the paper point to ensure that the primer has permeated to the complete depth of the canal preparation. *(c)* In the bonding step (3), a thin microbrush is used to apply the bonding agent, a paper point is used to remove excess bonding resin, and it is finally light cured.

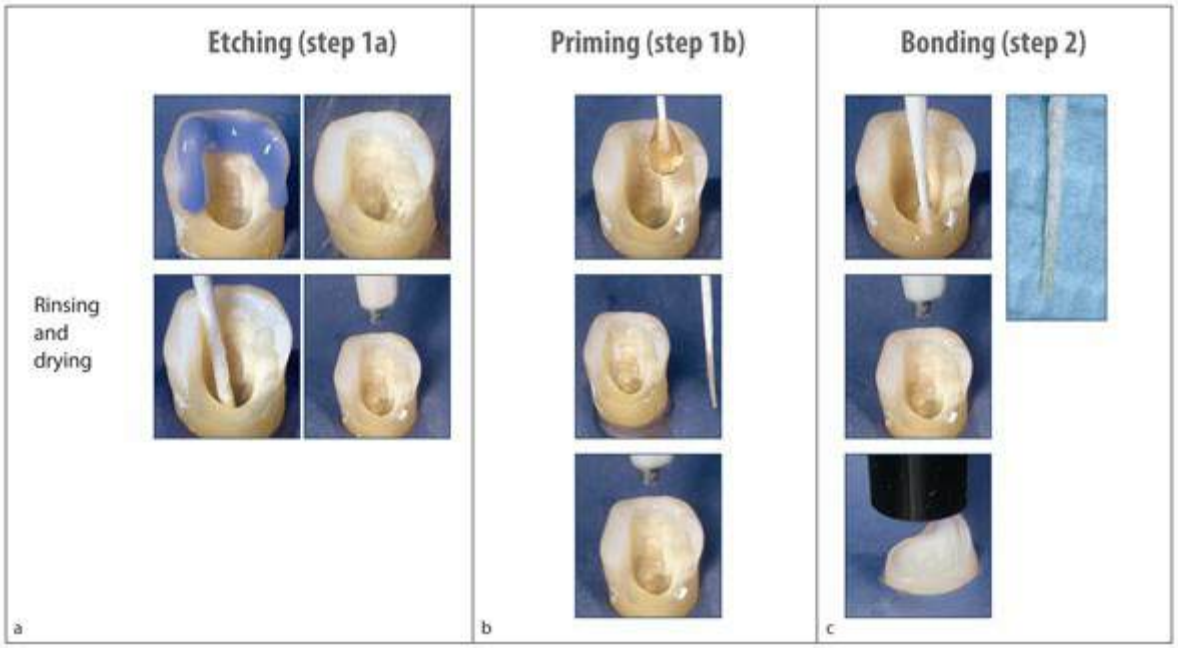


Fig 9-15 Step-by-step procedure of intraradicular bonding using a two-step self-etching adhesive. (Courtesy of Adamo Monari, Verona, Italy.) *(a)* In the etching step (1a), phosphoric acid gel is selectively applied to the enamel only. After rinsing, a paper point is used to remove water prior to priming. *(b)* In the priming step (1b), a thin microbrush is used to carry the primer to the full extent of the prepared canal, and a paper point is used to ensure adequate removal of excess primer/etchant prior to gentle air drying. *(c)* In the bonding step (2), a thin microbrush is used to apply the bonding agent, a paper point is used to remove excess bonding resin, and it is finally light cured.

The pulp tissue, predentin, mineralized dentin, and surrounding cement layer can be identified through analysis of the radicular substrate. *Predentin* is defined as the unmineralized organic matrix lining the innermost pulpal portion that is removed by pulpal tissue instrumentation and, in the case of post-space preparation, the use of calibrated burs. These root canal procedures create a dental substrate consisting of mineralized intraradicular dentin.[215,216] Although intraradicular and coronal dentin exhibit only minor morphologic and biochemical differences in substrate composition, significant differences have been found in the bond strengths to these different materials. Many studies also have shown that bond strength decreases from the coronal to the apical third of the root canal because of the significant reduction in the number of tubules toward the apical region and changes in the ratio of peritubular to intertubular dentin from the apical to the coronal third.[215,216] Although the greater proportion of intertubular dentin near the apex should result in higher bond strength in the apical third (versus coronal third) of the root, mineral deposits resistant to acid demineralization are frequently present near root apices and significantly affect the formation of the hybrid layer.[213]

These intrinsic substrate modifications result in reduced peritubular dentin infiltration and resin tag formation, compromising the achievement of high bond strength in the apical third of the root. The lower bond-strength values resulting from reduced adhesive impregnation reported by some authors[216] are likely because of a combination of these factors. Because hybrid layer thickness and resin tag morphology have recently been found to contribute only minimally to bond strength, the reduced bond strength to intraradicular dentin is likely because of factors other than substrate morphology, such as the clinical challenges posed by the limited endodontic space and the high cavity-configuration factor (C-factor).[217]

Clinicians would thus benefit from the development of a standardized, predictable, and reliable adhesive procedure for bonding within the endodontic space. To improve immediate bond strength and the long-term stability of the adhesive interface, operators also must be familiar with the primary factors that may negatively impact adhesion in this region and be able to critically evaluate clinical situations. Clinical factors (procedures and materials) and the geometric characteristics of the endodontic space can affect the use of adhesives and luting procedures in root canals.[213,218,219]

Because endodontic therapy seeks to disinfect the root canal space and fill it with an inert material, the creation of a tight seal with the tooth structure is mandatory. Adhesion to intraradicular dentin requires the creation of a perfectly clean post space because of the modification of intraradicular dentin caused by the use of sodium hypochlorite (NaOCl) rinses, ethylenediaminetetraacetic acid (EDTA), endodontic cements, gutta-percha, or other endodontic filling materials.[218]

Endodontic smear layers

The morphologic features, composition, and thickness of the endodontic smear layer are determined by endodontic instrument type, irrigation method, and tooth substrate type.[42] Whereas coronal smear layers contain materials derived from the dentinal matrix, endodontic smear layers contain a wider variety of inorganic and organic materials, including odontoblast fragments, microorganisms, and necrotic material.[220] The thickness of the smear layer ranges from 0.5 to 2.0 μm, with smear plugs obstructing dentinal tubule orifices and extending up to 40 μm into the tubules. Debris and pulp tissue remnants on dentinal surfaces within the root canal may prevent effective bonding; this endodontic smear layer forms a barrier that significantly affects adhesive bonds between instrumented canal walls and restorative materials and impacts the resin cementation of fiber posts. Although the

retention of the smear layer may be desirable in some adhesive dentistry applications, it should be removed in endodontic procedures.[221] Because the endodontic smear layer may harbor microorganisms and protect bacteria within dentinal tubules, removing it from infected root canals allows the penetration of intraradicular medications into the tubules.[222] Several studies have also reported that the removal of the smear layer promotes better adhesion of obturation materials and increases the penetration depth of sealers within the dentinal tubules by 10 to 80 μm.[220]

In addition to the smear layer produced by manual or rotary instrumentation of root canal walls, the subsequent preparation of the post space creates a thicker smear layer composed of debris and sealer/gutta-percha remnants that significantly influences fiber post adhesion.[223] In post-space preparation, the use of drills to remove the root-filling material produces this second smear layer. Drilling friction plasticizes the sealer and gutta-percha remnants contained within this layer, reducing the penetration and chemical action of fiber post bonding agents. Furthermore, the endodontic canal space allows only minimal irrigation. The achievement of optimal post retention thus requires the creation of clean dentinal surfaces after mechanical post-space preparation, particularly with the use of resin cements.[224]

Some authors have suggested that pretreatment with NaOCl and phosphoric acid[225] or a chelating agent can efficiently remove large areas of smear layer that would otherwise be unavailable for bonding and resin cementation of fiber posts, thereby improving post retention. The use of ultrasonic instrumentation and EDTA before bonding has also been suggested to reduce the presence of debris and open tubules.[226] Although these proposed pretreatments enhance the removal of the smear layer, the effectiveness of interfacial bonding is related primarily to the choice of bonding procedure.[227]

Clinical factors

Several clinical factors may affect adhesion to intraradicular dentin. The use of some disinfectants or medications during root canal preparation may adversely affect dentin-post bond strength,[228,229] as can bleaching and retreatment procedures.[218]

Irrigation of the endodontic space exposes radicular and coronal dentin to various disinfecting solutions that alter the dentinal surface and affect interactions with resin-based materials used for root canal obturation or coronal restoration.[213,230]

The effects of solutions such as EDTA and NaOCl on dentin have been investigated extensively.[229,231,232] Although NaOCl is usually recommended for endodontic irrigation, it has strong oxidizing properties and should not be used with resin-based materials. The use of NaOCl produces an oxygen-rich layer on the dentinal surface that can significantly reduce bond strength and increase microleakage.[231] Thus, the operator must employ clinical procedures that optimize adhesive-dentin and post-resin bond strength when using resin cements to lute endodontic posts. The use of a solution containing 10% ascorbic acid or 10% sodium ascorbate after NaOCl irrigation can improve adhesion to intraradicular dentin by completely reversing the compromised bond obtained with 5% NaOCl–treated dentin.[231]

Hydrogen peroxide (H_2O_2) may also be used during canal instrumentation to effectively remove pulp tissue remnants and dentin debris. However, the use of 3% H_2O_2 has been found to negatively affect the strength of bonds between resin cement and root canal dentin.[233] When it reacts with NaOCl, H_2O_2 breaks down and liberates water and oxygen, both of which may strongly inhibit the interfacial polymerization of resin-based bonding materials. The use of RC-Prep (Premier Dental Products) as a lubricant and demineralizing agent has also been found to reduce the strength of bonds between resins and root canal dentin. The polyethylene glycol vehicle (Carbowax, Dow Chemical) included as a lubricant in RC-Prep is difficult to eliminate completely during rinsing and inhibits resin polymerization. However, treatment with 10% ascorbic acid following the application of RC-Prep can restore bond strength to control values.[231]

Intraradicular irrigating solutions (NaOCl, H_2O_2, EDTA) directly affect the structural components of root canal dentin, significantly reducing microhardness.[232] Although these irrigants soften the dentinal walls, which may be clinically desirable because it permits rapid preparation and facilitates access to small, tight root canal spaces, adhesion and sealer performance are negatively impacted in chemically treated dentinal surfaces. The use of 0.2% chlorhexidine gluconate as an irrigating solution has been suggested to provide optimal obturation because it does not affect the microhardness and roughness of root canal dentin.[228]

Calcium hydroxide paste may be placed in the root canal between endodontic treatments for its antimicrobial properties and other beneficial effects. However, calcium hydroxide cannot be removed completely before obturation, and residual particles may form a physical barrier to bonding in some areas.[234] The high pH of calcium hydroxide may also neutralize the self-etching/primer solutions of self-etching adhesives, reducing etching effectiveness and bond strength.

Eugenol, which is included in some sealants and in zinc oxide–eugenol temporary cements, negatively affects resin polymerization and compromises bonding effectiveness because it can permeate dentin.[235] Like other phenolic compounds, eugenol is a radical scavenger that inhibits resin polymerization.[236] To avoid suboptimal polymerization and reduced bond strength, the application of a eugenol-based sealer or temporary material should be followed by the mechanical cleaning of root canal walls with detergent or alcohol to remove all residual material.[237] Such thorough cleaning may facilitate the removal of oily debris layers before bonding.[218,219]

During endodontic retreatment procedures, radicular and coronal dentin may be exposed to gutta-percha solvents (eg, chloroform, halothane) applied within the root canal.[238] These strong lipid solvents may alter the chemical composition of the dentinal surface and organic matrix, affecting the interaction with resin-based restorative materials. The solvents may be redeposited as a waxy film on the root canal surface, interfering with resin-dentin bonds and significantly reducing bond strength.[238] See chapter 21 for more on the restoration of endodontically treated teeth, including post preparation and cementation.

Bleaching

Tooth discoloration resulting from endodontic treatment is a serious problem that has received increasing attention in esthetic dentistry.[237] Although a variety of techniques can be used to improve esthetic outcomes, tooth whitening (bleaching) is the most conservative and cost-effective method of enhancing nonvital tooth appearance. However, bleaching has been associated with several side effects,[239] including reduced bond strength in resin-based restorative materials.[237] This reduction in bond strength is because of the oxygen-rich surface produced by H_2O_2-based products, which significantly inhibits adhesive polymerization.[240] Bond strength may be improved by placing the final restoration 1 week after a bleaching procedure to allow the release of residual oxygen. Extended polymerization times may partially mitigate polymerization inhibition when immediate bonding is necessary.[240]

Incompatibility between simplified adhesives and chemically/dual-cured composites and resin cements

The two-step etch-and-rinse and one-step self-etching adhesives have been found to be incompatible with chemically or dual-cured composite materials.[241,242] This results in reduced bond strength in inverse proportion to adhesive acidity. In addition, adverse chemical interactions occur between sub-optimally polymerized acidic adhesive resin monomers and catalysts in the composites.[242] These adhesives exhibit incompatibility with such composites because of the increased concentrations of lower-pKa acidic resin monomers that are included in these products to increase permeability.[243] These monomers can react with the basic components (aromatic tertiary amines) of the composite, inhibiting polymerization and/ or creating a hypertonic environment in which fluid is drawn osmotically from bonded hydrated dentin through the permeable adhesive layer.[243]

The use of an intermediate resin layer in three-step etch-and-rinse and most two-step self-etching adhesive systems (ones that do not contain an acidic monomer in their adhesive resin components) renders them compatible with chemically or dual-cured resin composites.[241] The resin layer is less acidic and less hydrophilic, which prevents the negative acid-base reaction. The layer also prevents direct contact with the acidic monomer components in the primer layer and reduces the permeability of resin-tooth interfaces.

Dual-curing adhesives are commonly used for bonding to root canal dentin because they can self-polymerize without light in the deep regions of the post cavity. However, the incompatibility between one-step self-etching adhesives and chemically/dual-cured composites precludes their use for indirect bonding in areas that are inaccessible to light activation. The use of one-step self-etching adhesives should thus be avoided in these situations, although the acid-base reaction may be overcome with the use of a ternary catalyst.[243,244]

Geometric factors

The endodontic use of methacrylate-based resin materials is affected by polymerization-associated shrinkage stresses that are great enough to cause debonding from dentin.[245] This debonding may result in reduced retention and increased leakage with the enhanced potential for bacterial invasion at the tooth-resin interface.

Feilzer et al[246] in 1987 found that shrinkage stress was related to the *C-factor*, defined as the ratio of bonded to unbonded surface area in the restoration. High C-factor values are indications of situations where the material is polymerizing under

greater external constraint, thus establishing high internal stresses, which may exceed the bond strength to the tooth. The viscoelastic properties of the resin material also affect shrinkage stress; rigid resins produce high shrinkage stresses that have a greater potential to increase gap formation at the tooth-resin interface.[247] Bonds may be preserved if the elasticity of the surrounding materials is sufficient to relieve shrinkage stress caused by the wall-to-wall contraction of the resin material.[21] However, such relief is unlikely in deep root canals because of the geometry of the cavity and the thickness of resin films.[248] Root canal cavities have small unbonded surface areas that provide insufficient stress relief, increasing the probability of debonding in one or more bonded areas. Braga et al[249] found that shrinkage stress in composite restorations was affected more strongly by cavity depth than by diameter, which has negative implications in endodontic cavities. Negative geometric factors affect the cementation of endodontic posts to prepared post spaces, significantly compromising the ability to create leak-free interfaces. The C-factor in post spaces may exceed 200, whereas coronal restorations are characterized by C-factors of 1 to 5.[245,250]

Rheologic properties of the luting materials are also important during the luting procedure. In particular, a high percentage of filler in the resin cement has been found to significantly decrease bond strength because of increased polymerization stress at the interface.[217]

Evaluation of Bond Strength

The in vitro performance of adhesive systems is most frequently evaluated using macro- or microbond strength tests, the distinction being dependent on the size of the bonded area (Fig 9-16). Because dental research literature is replete with bonding studies, a brief review of the types of studies that will be encountered, as well as their primary strengths and weaknesses, will be provided.

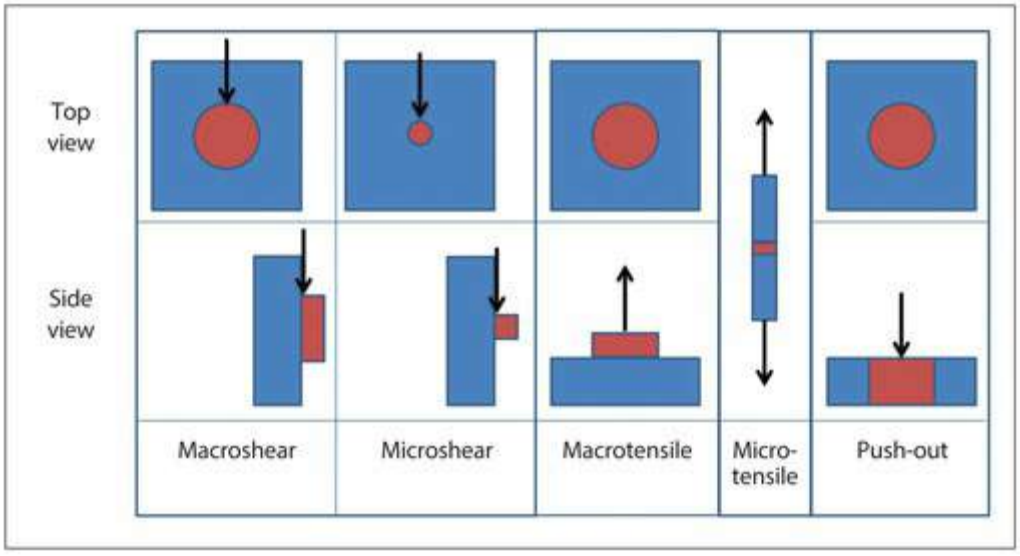


Fig 9-16 Schematic setups of the main bond-strength tests (*left to right*: macroshear, microshear, macrotensile, microtensile, push-out).

Macrobond strength tests

Bonded areas exceeding 3 mm^2 are assessed by macrobond strength tests[113] employing a shear, tensile, or push-/pull-out technique. Bond-strength values obtained using macrobond tests are usually lower than those obtained using microbond tests (area ~ 1 mm^2) because the increased surface area has a higher probability of containing large flaws that lead to bond failure[113,251,252] (see Fig 9-16).

Macroshear tests

In macroshear tests, a composite cylinder of restorative material is fixed to a flat enamel or dentinal surface after the application of adhesive (see Fig 9-16). Force is then applied parallel to the tooth surface until the bond fails.[251] Although loops, chisels, or knife edges have been used for load application, the use of a guillotine jig is recommended in the International Organization for Standardization (ISO) standard no. 11405.[253] Differences among studies in force-application devices as well as crosshead speed often prevent the direct comparison of test results.[252]

Although macroshear tests are the most commonly used method of assessing bond strength because of the simplicity of specimen preparation,[254] the stress generated at the bonding interface in such tests may lack uniformity. Such uneven stress may significantly affect the mode of bond failure.[255–258]

Macrotensile tests

In macrotensile tests, tensile force is applied to a cylinder of restorative material (typically composite) that has been polymerized onto a flat enamel or dentinal surface after the application of adhesive.[259] Although such tests are used less commonly than macroshear tests, they allow the evaluation of bond strength in luting agents used with hard materials such as metal alloys and ceramics.[260,261]

Push-/pull-out tests

In push-out tests, a specimen is embedded into a thin slice of material with its long axis oriented in the plane of the slice (see Fig 9-16). A compressive axial load is then applied to the upper surface of the specimen, extruding it through the base of the slice until debonding occurs.[262] The retention of posts luted in root canals[263,264] is typically assessed with push-out tests. This method also allows the concurrent microscopic analysis of marginal adaptation in the same specimen.[251]

The pull-out test is usually applied to fiber-reinforced composites, by which a single fiber, partially embedded within a resin matrix, is extracted under a tensile load.[265]

Microbond strength tests

Bonded surfaces with areas of 1 mm^2 or less are assessed using microbond strength tests, which may be divided into microshear and microtensile tests (see Fig 9-16).

Microshear tests

Microshear tests evaluate bond strength using restorative material buildups with small ($\leq$ 1 mm^2) cross sections of bonded areas. Specimen preparation for these tests is relatively simple, and multiple specimens can be prepared from a single tooth. However, it is impossible to confine the adhesive to the test area, as specified by ISO standard no. 11405 (2003). Furthermore, the distribution of stress in microshear tests is concentrated nonuniformly in the substrate, leading to the significant underestimation of stress. These shortcomings limit the recommended use of microshear tests to brittle substrates that are susceptible to fracture during specimen preparation, such as enamel or glass ionomers.

Microtensile tests

Bond strengths to dentin are most frequently evaluated using microtensile tests (μTBS). In these tests, large restorative material buildups are placed on a flat dentin surface after the application of adhesive. Multiple specimens may then be obtained from each tooth by removing 1 × 1–mm or smaller sections (ie, sticks) from the tooth-composite preparation. The specimens are then loaded to failure, and the tensile load at failure is divided by the cross-sectional area of the bonded interface to determine microtensile bond strength. Specimens may be prepared using trimming or nontrimming procedures[266,267]; the *nontrimming* technique is simpler and requires less preparation time. *Trimming* procedures create dumbbell- or hourglass-shaped specimens by trimming them at the interface. Although this technique improves the concentration of stress at the interface, specimen preparation is complex and highly operator dependent. The use of gripping devices or glue with specimenjig attachments, the speed of loading, and the alignment of specimens may also affect the results of microtensile tests.

Pretest failure occurs frequently during the preparation of specimens for microtensile tests. Several approaches to the consideration of such failures have been proposed, although each is associated with an analytic shortcoming. The exclusion of all pretest failures from statistical analyses may lead to the overestimation of bond strength, whereas the designation of a 0-MPa bond strength value to each failure may result in the underestimation of bond-strength. A predetermined bond-strength value, such as the lowest measured value within the test group, may also be assigned to each pretest failure.

Hybrid Layer Aging and Interface Stability

Because the dentin substrate is impregnated by resin monomer blends during bonding, the stability of a bonded interface depends on the creation of a compact and homogenous hybrid layer.[22] Bonding with etch-and-rinse systems includes preliminary etching to demineralize the substrate and create a porous surface, followed by the application of bonding monomers.[33,34,37] The complete infiltration of the adhesive into the substrate is necessary to achieve a stable bond with these systems.[16,34,66,68,268,269] In self-etching adhesive systems, acidic co-monomers simultaneously demineralize and infiltrate the dentin; interface stability thus depends on effective co-monomer–substrate bonding.[35] Mild (pH ~ 2) two-step self-etching systems also produce chemical bonds between carboxyl or phosphate groups contained within functional monomers and residual hydroxyapatite crystals on the

dentinal collagen scaffold; these bonds synergistically enhance long-term stability.[24,35] However, there is also some evidence of continued etching after curing for some aggressive one-step self-etching systems,[45,161] which would lead to exposed collagen and accelerated bond degradation.

Physical and chemical factors contribute to the clinical longevity of the hybrid layer. Physical factors affecting interface stability include occlusal chewing forces and repetitive expansion and contraction stresses produced by temperature changes within the oral cavity.[179,270] The interface is also affected by bacterial and acidic agents in dentinal fluid, saliva, food, and beverages, which produce various patterns of collagen fibril and resin degradation[16,23] and resin-monomer elution (likely because of suboptimal polymerization).[109,115]

The hybrid layer is created by the combination of the dentinal organic matrix with residual hydroxyapatite crystallites, resin monomers, and solvents; individual components may be affected by aging, and degradation may result from the synergistic combination of several components within the hybrid layer.[22]

Resin degradation

After storage, the hybrid layer may exhibit two patterns of degradation that compromise the strength of the resin-dentin bond: collagen fibril disorganization and the hydrolysis of resin from interfibrillar spaces within the hybrid layer.[268]

Hydrolysis is a primary means of resin degradation within the hybrid layer.[22,112] This chemical process involves the disruption of covalent bonds between polymers when water is added to ester bonds, resulting in the loss of resin mass[112,271] and contributing to the reduction in long-term bond strength created by dentinal adhesives.[16,22] In efforts to reduce such degradation, several studies have investigated the degree of water sorption within the hybrid layer produced by recently introduced simplified adhesive products.[15,16,22,23] Two-step etch-and-rinse and one-step self-etching adhesives employing hydrophilic acidic resins have shown high water sorption.[15] Water sorption significantly decreased the elastic modulus of resins, as plasticization of the polymer contributes to reduced bond strength in a manner similar to but independent of resin hydrolysis.[23,272]

Because hydrolytic degradation occurs only in the presence of water, it is closely associated with the hydrophilic properties of adhesives and the degree of water sorption.[112,273] The combination of hydrophilic and ionic resin monomers in two-step etch-and-rinse and one-step self-etching adhesive systems thus produces a

bonded interface that lacks a solvent-free hydrophobic resin coating.[112] Hybrid layers created using such systems behave as semipermeable membranes that permit the movement of water throughout the bonded interface, even after adhesive polymerization[114] (Fig 9-17).

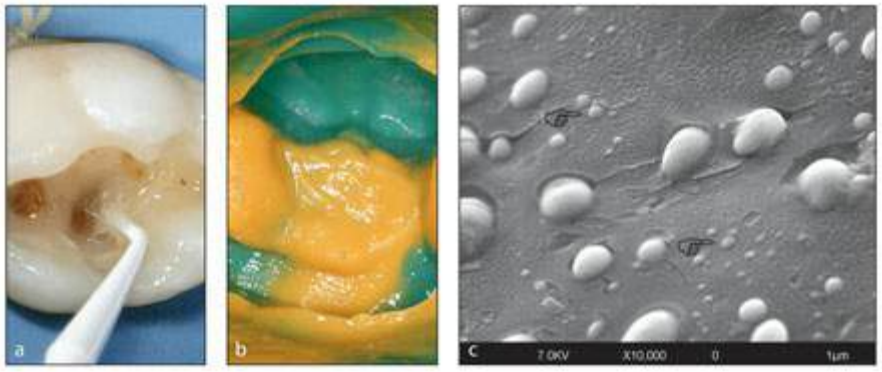


Fig 9-17 Illustrative steps of the in vivo analysis of the permeability of adhesives. A cavity was prepared and bonded *(a),* and an impression of the cavity floor was obtained *(b)*. After a cast with epoxy resin was poured, specimens analyzed under FE-SEM revealed water droplets emanating from the adhesive surface *(c)*. These droplets are the morphologic evidence of water that seeped from the adhesive layer during the setting time of the hydrophobic impression material, forming major droplets as well as minor droplets *(pointers)* over the adhesive. These droplets may compromise coupling between the adhesive and the resin-based restorative material. (Reprinted from Breschi et al[22] with permission.)

Etch-and-rinse adhesive systems are prone to nanoleak-age because of insufficient adhesive impregnation along the resin-dentin interface after demineralization.[65,67,177] The higher hydrophilic monomer content of two-step etch-and-rinse adhesives (versus three-step systems)[112] results in greater permeability after polymerization, which facilitates moisture uptake and increases nanoleakage expression.[114] Although all types of adhesive systems have shown variable degrees of incomplete polymerization associated with permeability to fluid movement,[109,115,274] these issues are more pronounced in simplified adhesives (eg, two-step etch-and rinse, one-step self-etching systems), likely because of a higher content of hydrophilic monomers[109,115,274] (Fig 9-18). Partially cured adhesives exhibit greater permeability, which may increase water sorption and compromise the long-term integrity of adhesive-composite bonds.[124] In contrast, more complete polymerization (and thus reduced permeability) can be achieved with the use of dentin bonding systems that include solvent-free hydrophobic monomers.[115]

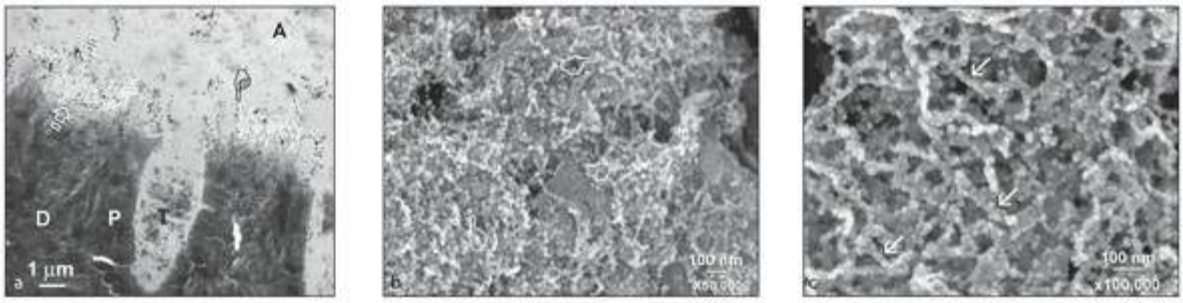


Fig 9-18 Correlative transmission electron microscope (TEM) and FE-SEM examination of nanoleakage produced by a one-step self-etching adhesive. The adhesive was applied to the dentin passively (without agitation). *(a)* The TEM micrograph of the resin-dentin interface shows the presence of a 2 µm–thick hybrid layer *(white arrows)*. The entire hybrid layer is filled with fine, isolated silver grains that were probably affiliated with the partially demineralized, unstained collagen fibrils *(pointer)*. Very fine silver grains could be seen within the adhesive layer. A—adhesive; D—dentin; P—peritubular dentin; T—tubule impregnated by the adhesive. *(b)* FE-SEM images taken from a region within the fractured hybrid layer show that the latter was completely covered with fine silver grains, and *(c)* at higher magnification these fine silver grains were in the range of 5 to 20 nm in diameter. These silver grains were affiliated with the dehydrated and shrunken collagen fibrils. (Reprinted from Suppa et al[275] with permission.)

Collagen fibril degradation

Bond longevity may also be negatively impacted by increased water content at the bonded interface resulting from resin and collagen degradation. In addition, the degradation of collagen is caused primarily by the presence of water. Resin or collagen hydrolysis may degrade the hybrid layer, causing the loss of resin from interfibrillar spaces and the disorganization of collagen fibrils[268] and physically weakening the resin-dentin bond.[23] After long-term storage, bonded materials have exhibited resin elution and/or hydrolytic degradation of collagen matrices.[22,23] In vitro and in vivo studies have detected deterioration within the hybrid layer that indicates the presence of numerous exposed collagen fibrils.

The dentinal surface is often incompletely hybridized in resin-dentin bonding, regardless of the adhesive system used. Such incomplete hybridization may leave exposed collagen fibrils and residual adhesive solvent and/or surface water that increases vulnerability to hydrolytic degradation.[101,119,276]

Dentin bonding may also be affected by the breakdown of collagen matrices by host-derived proteinases released during the development of dentinal caries and periodontal disease.[277,278] The presence of matrix metalloproteinases (MMPs) in human dentin can be inferred based on evidence for colla-genolytic/ gelatinolytic activities in partially demineralized collagen matrices.[279–282] *MMPs* are zinc- and calcium-dependent endopeptidases that are incorporated within the mineralized

dentinal matrix during tooth development.[283] When released and activated during dentin bonding procedures, these endogenous enzymes can degrade extracellular matrix components[283] (Fig 9-19).

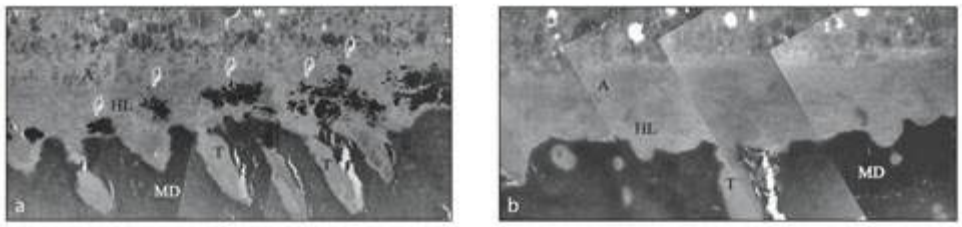


Fig 9-19 TEM images obtained combining numerous micrographs of a representative specimen treated without *(a)* or with *(b)* 0.2% chlorhexidine for 30 seconds, then bonded with a two-step etch-and rinse adhesive (Adper Scotchbond 1 XT) and stored for 2 years in artificial saliva at 37°C. *(a)* The control adhesive interface reveals extensive interfacial silver nanoleakage *(pointers)* due to individual silver grains and large clusters of silver deposits within the collagen fibrils of the hybrid layer. *(b)* If chlorhexidine was used, the adhesive interface revealed only very few scattered particles of silver nanoleakage within the hybrid layers. Bar = 2 µm. HL—hybrid layer; MD—mineralized dentin; T—dentinal tubules; A—filled adhesive. (Reprinted from Breschi et al[85] with permission.)

In vitro studies have found that MMPs thin or remove collagen fibrils from incompletely infiltrated hybrid layers in aged, bonded dentin,[85,87,89,284] and in vivo studies have observed MMP-induced collagen degradation in basal hybrid layers.[88,89] The use of chlorhexidine, an antibacterial agent with MMP-inhibiting properties, on acid-etched dentin during etch-and-rinse bonding procedures can maintain hybrid layer collagen integrity.[85–88] This effect confirms the indirect involvement of MMPs in collagen degradation (see Fig 9-19).

Cysteine cathepsins, which can be activated in mildly acidic environments, may also contribute to collagen degradation.[285] The activation of dentin-bound cathepsins may also activate matrix-bound MMPs, increasing long-term hybrid layer instability.[16,22]

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